PIXEL

A Competent Book on Computer Education with AI + ML

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(M. Tech.)

5

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I choose **JPEN Books** because ...

- **JPEN Books** are based on an Interactive Technique that makes students Highly Motivated to gain confidence in their own Abilities.
- **JPEN Books** use conceptual learning in a step-by-step method.
- **QR Codes** are provided for every chapter to unlock the very Fine Digital Content.

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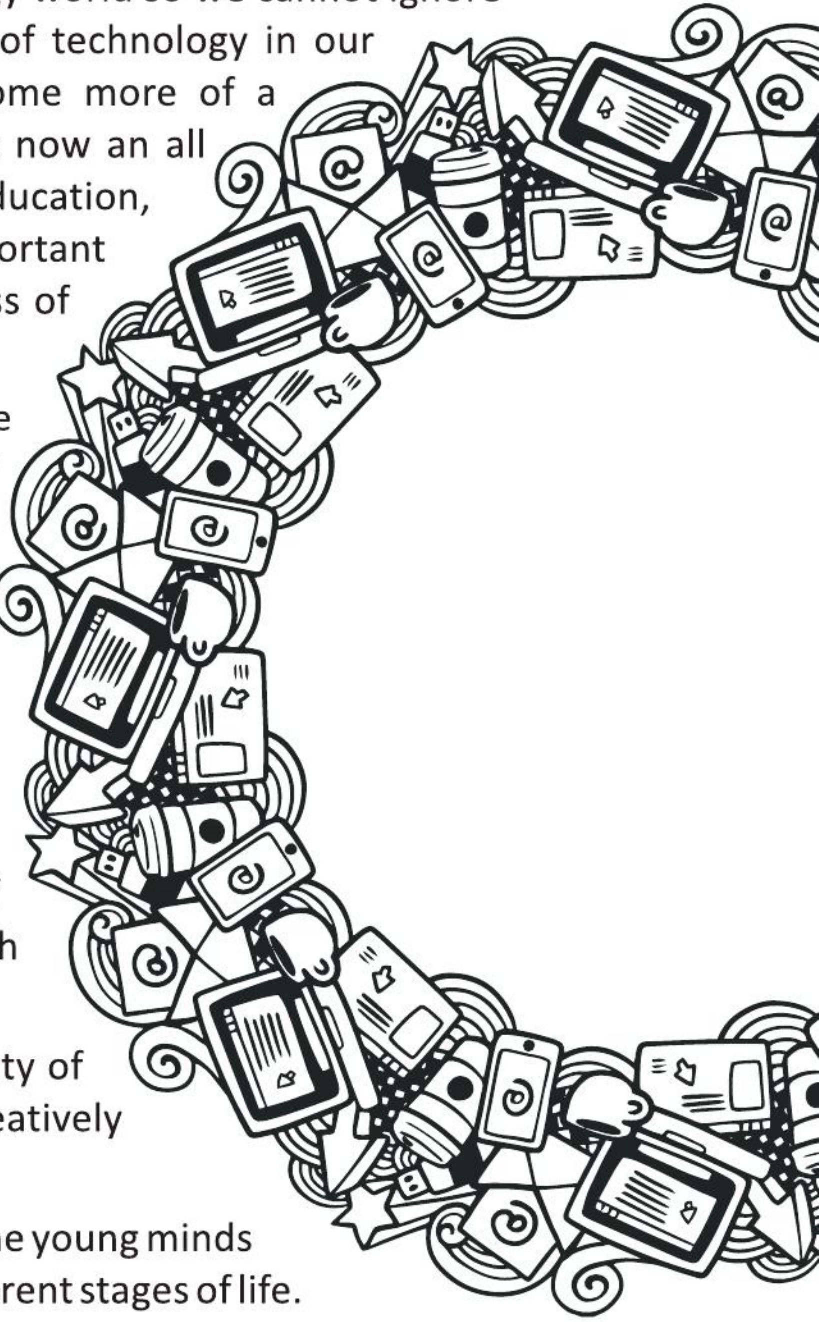
Preface



Today, we are living in the Digital technology world so we cannot ignore the importance of computer. The impact of technology in our modern-day life is such that it has become more of a necessity for us rather than a facility. It is now an all invasive component of our life, be it an education, occupation or our common life. It is an important tool that adds massive value to the process of teaching and learning.

Our series **Pixel** is strictly based on the guidelines given by **National Education Policy (NEP)** and **National Curriculum Framework (NCF)**. Young children need to be given the tools and techniques to learn, master and integrate the use of computer in their learning life from the beginning.

While creating this series, we took special care of few things that happen in the world of Digital technology, latest updates of software versions, and an all-new approach to bridging the digital learning.

This is an activity-oriented series with plenty of screen captures, colourful illustrations, creatively designed exercises and activities.

We sincerely hope that our books will help the young minds to equip them with the required skills at different stages of life.

Any suggestions for the improvement of these books will be highly appreciated.

—Author and Publishers

INCORPORATES NEP 2020 AND NCF 2023

Created with a new dimension of joyful learning along with completely mapped parameters of National Education Policy, 2020

01 Critical Thinking
Strengthening decision-making and innovation

02 Art Integration
Promoting art-based learning

03 Creative Learning
Thinking out of the box

04 Subject Enrichment
Aimed at understanding and skill developing in specific subject

05 Inter Disciplinary
Connects knowledge across the curriculum for holistic development

06 Computational Thinking
Enables to help solve problems and make thinking more engaging.

07 Experimental Learning
Encourages learning by doing through hands-on activities

08 Life Skills and Values
Inclined towards empowering moral values in learners

09 Communication
Emphasis on soft skills and power of language

SYNOPSIS

Outline includes the topics to be learnt in the chapter.

Outlines

- Computer : An Electronic Machine
- Features of a Computer
- Types of a Computer

Info Zone include some facts related to chapter to enhance the knowledge of the students.

Info Zone

The first version of MS Paint was introduced with the first version of windows in 1985.

Smart Tip provides keyboard shortcuts for menu commands, to help users save time while performing routine operations.

Smart Tip

You can also open Paint by typing Paint in the Search Box next to the Start button. Then, click on the Paint button.

Computer Etiquette presents computer ethics and manners to be followed while working on a computer.

COMPUTER ETIQUETTES

While working on a computer, keep your neck relaxed. Practise this exercise to avoid any strain on your neck.

Let's Implement contains carefully graded exercises to test the knowledge of concepts learnt in a chapter.

Let's Implement

A Tick (✓) the correct option.

1. A monitor is also called :
   (a) LCU ◯ (b) VDU ◯
2. Which part of a computer is called the brain of a computer?
   (a) CPU ◯ (b) VDU ◯
3. It is used to give the output of our work on paper.
   (a) Speakers ◯ (b) Printer ◯

Refreshment Corner includes interesting exercises for enhancing the skills with fun.

Refreshment Corner

A Colour the alphabet keys that spell your name with yellow.

Practical Session for practical hands-on exercises to strong then the concepts learned by doing.

Practical Session — Experiential Learning

- Visit your computer lab and open Paint. Draw and colour the following.

Racking Your Brain includes the life skills and values questions.

RACKING YOUR BRAIN — Life Skills and Values

Smart devices are used to make our tasks easier. Should we always use them at small tasks? Justify your answer.

Just Have a Discussion stimulates a small discussion for the topics being covered in the chapter which in turn improves the communication and logical skills.

Just Have a Discussion — Communication

Discuss the different components of ScratchJr in class in detail.

Teacher's Note provides important information and suggestions on creative approaches to a chapter or a topic.

TEACHER'S NOTE

- Give enough exercises to students so that they get practice of using the tools of Paint as well as using the mouse.

Contents

1

WORLD OF COMPUTER

Outlines

- Uses of Computer
- Computers in Different Fields

Uses of Computer

Nowadays, computers are used for the smallest of things to the large ones. They perform various tasks with maximum accuracy in the minimum time.

They've been useful in every field like health, education, businesses, construction, and whatnot. If there were no computers, it'd have been difficult to manage all the tasks at the same time.

Computers are used in different fields because of their characteristics and functionality, such as, business, household, offices, educational institutions, engineering etc. Computers have taken the business and technology industry to a whole new level.

Computers are a necessary component of our daily lives in the current day. This means that computers are used in practically every industry, which facilitates and expedites our daily tasks. Today, computers are used in various locations, including our homes, banks, stores, schools, hospitals, and railways. We must be aware of the fundamentals of computers because they are such an integral part of our daily life.

At Home

At home, we usually have personal computers or laptops. They are used for :

- Playing music and movies.
- Playing games.
- Performing creative activities like drawing, designing, and making birthday cards.

- Sending e-mails and messages.
- Writing project work.
- Storing information like important dates.

With the help of an internet connection, a personal computer helps us to explore the world.

In School

In schools, a computer is used by the teachers, the office staff and the students.

- The teachers use it for preparing assignments, teaching aids and teaching of different subjects etc.
- The office staff uses computers for preparing results and grade cards, maintaining admission records, sending notifications to parents and keeping track of all the school activities.
- Students in school take lessons on computer, learn various subjects and gain valuable general knowledge.

Info Zone

William Higinbotham created the first video game called Tennis for Two.

In Business

Computers increase the productivity of a business and help them sustain in a dynamic environment. In businesses, computers are used to store and manage accounts and personal data, track inventory status and make presentations. Computers are favourable in doing transactions because they are more accurate and faster than humans.

In Hospitals

Nowadays, almost all the hospitals are well-equipped with computers. Various sections of hospitals using computers are the hospital office, operation theaters, laboratory, diagnostic section and the patient check-up area. The various functions performed are :

- The hospital office maintains records about patients, childbirth and death, records about hospital staff, doctors etc.
- Various test reports, diagnostic equipment like digital x-ray, ultrasound, MRI, etc. are computer-based.

- Medical research trials and disease detection are computerized.
- Dentists and eye specialists use computer-based equipment to do routine checkups.

In Education

Computers are broadly used in the education industry. They help with getting different materials like e-books, PDF, videos and what not. Computers have made education easy and yet effective. Students can take classes globally with the help of computers.

In Banks

In banks, computers play a vital role. The types of computer used in banks are networking computers, which are connected to a mother computer or a master computer. Banks use powerful computers for bulk processing and storage.

The various functions of computer in a bank are :

- Maintaining account details of account holders.
- Giving online facilities to customers for money transfer and payment of bills.

In ATM

Automated Teller Machines, or ATMs, are specialised computers that make it simple to manage the money in a bank account. Users can check their account balances, withdraw or deposit funds, print a statement of account activities or transactions, and even purchase stamps. Some uses are:

- The most common applications for an Automated Teller Machine are cash withdrawals and balance checks.
- Customers can now conduct financial transactions whenever it suits them. Today, banks have placed their ATMs in public areas such as highways, shopping centres, markets, hospitals, and airport terminals.
- Automated teller machines allow access anywhere, at any time.

In Designing

Computers are used in designing as follows :

- Architects design buildings and houses on a computer.
- Engineers design cars, airplanes, and many other machines on a computer.
- A dress designer designs dresses on a computer.

Computer-Controlled Cameras

The movements of the camera are controlled by the computer to capture images precisely.

- Doctors use miniature cameras to look inside the human body for diagnosing different diseases.
- Many banks, stores, and schools use security cameras to keep a watch on people's activities and movements. This helps to catch theft in banks or stores.
- Cameras on highways show traffic patterns. They help in monitoring speeding vehicles and the vehicle that does not follow other traffic rules.

At Shops and Multiplexes

Computers are used in shops and multiplexes for here for :

- Billing and item details.
- Ticketing and seat booking in multiplexes.
- Stock maintenance.

At Airport and Railway

At Airports and Railway stations, a computer helps in :

- Ticket booking and making reservations.
- Keeping records of arrivals and departures.
- For proper signalling.
- Airport have a special junction called the Air Traffic Controlling Office; which monitors proper landing and take off, weather monitoring and signalling.

Info Zone

A smartphone is a cellphone that allows you to do more than just make phone calls and send text messages.

Computers in Different Fields

In Entertainment Industry

- The workstation computers are leading a boom in the music industry. They have digitalized the music world.

- Animation and 3 D effects all have been developed through computers. You enjoy your favourite cartoons with the help of the computer graphics only.
- The special effects in movies like action movies, horror movies and thrillers are provided by computers.

For Defense and Safety

- Computers are used for vigilance purposes too. Security computers are installed in banks, public places, sensitive and secret areas to keep an eye on suspicious activities.
- Computers help in the operation of highly technical defense warplanes, specialized weapons and arms.

- At traffic signals, computers monitor speed and timely signalling.

Other Areas

The list of areas of computer usage is endless. Some other areas which use giant super computers are :

Space Research and Technology

In space science, computerised technology handles satellites and rockets launching systems and shuttles.

In Book Publishing

Computers help in creating colourful and attractive books.

- The books you are reading are also designed on computer by using book publishing software.
- Even, you can also create and design pages of your notes on your computer.

In Making Films

Earlier, creating animations used to be a difficult task as each picture in the sequence had to be drawn by hand. Today, computers have made it easy to make cartoon films and animations. Animation is a technique of giving a film or a character the appearance of a movement.

COMPUTER ETIQUETTES

1. Just like in real life, treat others online with respect.
2. Be carefully about the information what you share online?

Let's Implement

A Tick (✓) the correct answer.

1. We may use the computer at home :

   a. to create music and listen to music ◯   b. to cook food ◯

   c. to go from one place to another ◯   d. Both a. and b. ◯

2. Banks use a computer to :

a. keep a record of all accounts ◯ b. draw and colour images ◯

c. take photographs and edit them ◯ d. None of these ◯

3. We can use a computer for :

a. collecting information on projects ◯ b. launching rockets ◯

c. weather forecasting ◯ d. Both ii. and iii. ◯

4. Computers are used at railway stations and airports for :

a. proper landings and take-offs ◯ b. ticketing and reservation ◯

c. signalling ◯ d. All of them ◯

5. We may use computers at :

a. post offices and supermarkets ◯ b. banks and railway stations ◯

c. shops and schools ◯ d. All of these ◯

B Fill in the blanks.

1. At the railway station, a computer is used for printing ________________ .
2. Computers help ________________ in diagnosing many diseases.
3. ________________ are machines controlled by computers that help explore different planets.
4. ______________ that are linked to computers provide us information for space research.
5. Computers help us to withdraw cash from ________________ .

C Match the following.

1.	Animation	a.	ATM machine
2.	Computer lab	b.	Amitabh Bachchan
3.	Doctors	c.	Tom and Jerry
4.	Film	d.	medicine
5.	Bank	e.	microscope

D Write **True** or **False**.

1. You can get any type of information on the Internet. ☐
2. Computers are used to store patient data. ☐
3. Shops and supermarkets use programs to calculate the bills. ☐
4. All shops use computer program to list the record of their customers. ☐

E Define the following terms.

1. ATM ______________________________
   ____
2. Film ______________________________
   ____
3. Animation ______________________________
   ____

F Answer the following questions.

1. List five ways in which computer may be used at home.
2. How are a computer used in schools?
3. Give one way in which a computer is helpful to doctors.
4. Mention any three uses of a computer in science and technology.
5. What is an ATM? How is it useful to people?

Refreshment Corner

- **Read the names of the places given below and write Yes/No, whether you have seen a computer at the mentioned places or not. And if Yes, then what was the computer being used for :**

	Yes/No	Purpose
Post Office		
Bank		
Railway Booking Office		
Kitchen		
Movie Hall		
School		
Park		
Amusement Park		
Hotel		
Small Shop		

- **Project :**

1. What is net banking facility? Make a flow chart of how your parents login to their account?
2. Collect information about your classmates and prepare an Excel sheet of it. Store the data in your computer.
3. Search the Internet and various sources and find out about various satellites which are helpful in our life in many ways. Make a project.

RACKING YOUR BRAIN

Life Skills and Values

Tell the students some more uses of computers in school computer lab.

Teacher's Note

What will happen if all the data from your teacher's computer gets deleted and she/he does not have any backup?

2

MORE ABOUT WINDOWS 10

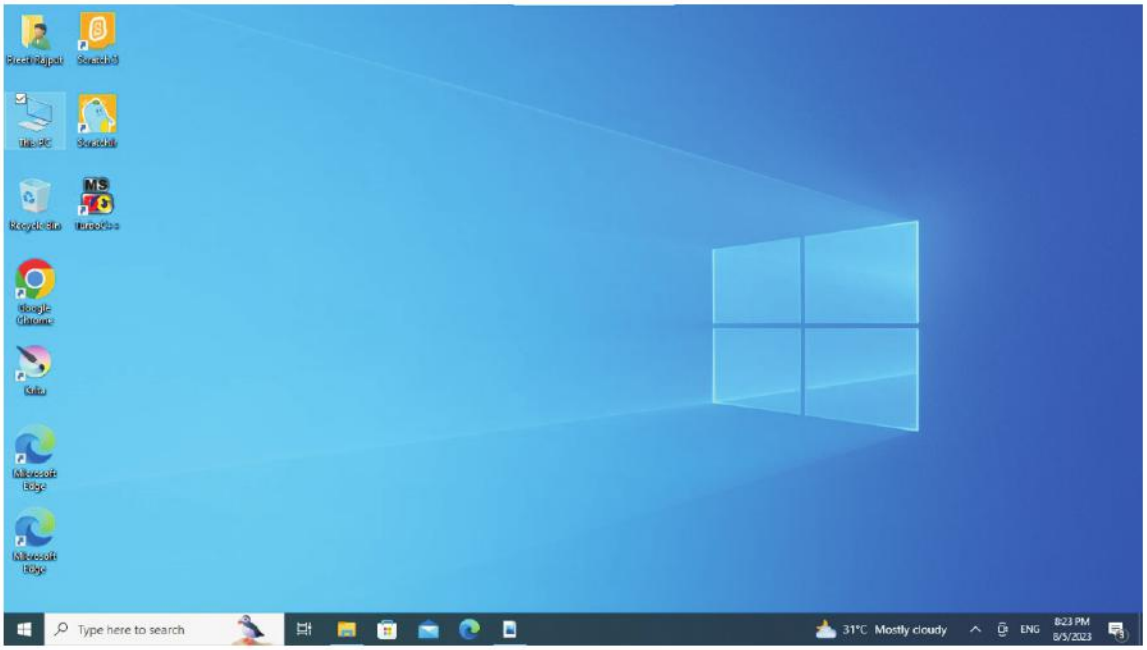
Outlines

- Booting
- Icons
- Displaying Multiple Windows at a Time
- Taskbar
- Live Titles

Booting

A computer needs an operating system to operate. When we switch on a computer, it takes some time to start. This happens because the operating system of the computer is loading in its memory. This process is called booting.

Windows is an operating system that acts as an interface (Graphical User Interface) between the user and the hardware. Once the process of booting is over, we see many small pictures on the monitor. These pictures are called icons.



The window that appears on the monitor when we switch on a computer is called a desktop. The desktop is the place where we find various icons.

Taskbar

A long horizontal bar present at the bottom of the desktop is called Taskbar. The Start button is present at the bottom left corner of the Taskbar. When we click on the Start button, a list of options is displayed. Next to the Start button is the Search box, which allows searching both the computer and the web. To the right of the Search box, is the Task View button which allows to create multiple desktop at the same time. The middle section of the Taskbar shows the programs that are open. The Notification Area is seen on the right side of the Taskbar. It includes utilities like Clock, Network Connectivity, Battery and Volume.

Info Zone

To change Date and Time, right click on the clock shown on the Taskbar and select adjust date/time.

Icons

The small pictures on the desktop are called icons. Some commonly used icons on the desktop are described below:

This PC

It helps us to select and work in different drives and folders that store information inside a computer.

Internet Explorer

It enables us to connect to and browse the Internet.

Recycle Bin

It stores the deleted files and folders till they are permanently deleted. It acts like a waste paper basket.

Network 

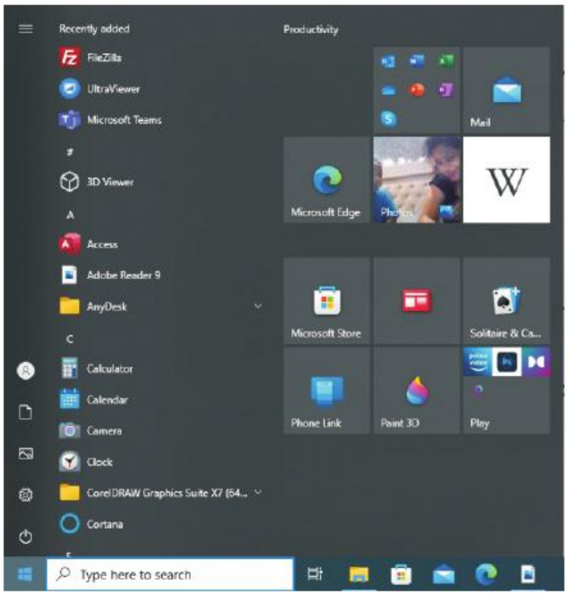
It allows us to access the shared files or devices on different computers in a networking environment.

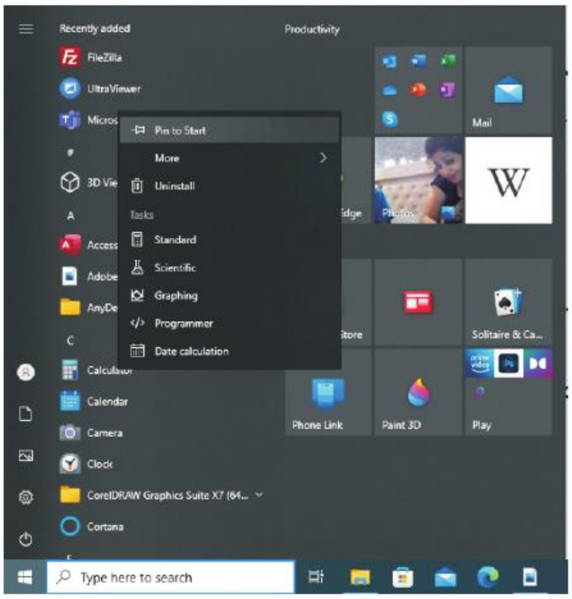
Shortcut Icons 

It provides a direct route to a specific application, document or a folder.

Live Tiles

Windows 10 displays the most popular universal apps in the form of resizable live tiles in the right pane of the Start menu. These tiles display the latest updates such as news and weather report. You can pin the desired tiles to the Start menu.





Follow these steps to pin tiles to the Start menu:

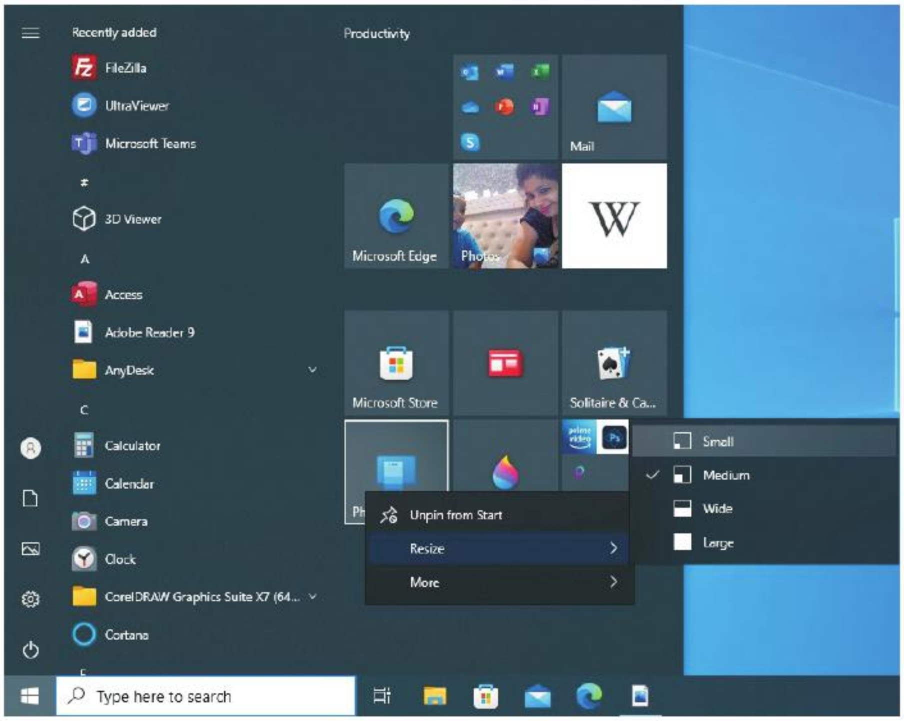
1. Click on the Start button.
2. Right-click on the app that you want to add to the 'Start' menu.
3. Select the option Pin to Start. The selected app is displayed on the right pane in the form of a tile.

Resizing Tiles

Tiles are available in four different sizes: small, medium, wide and large.

Follow these steps to resize a tile:

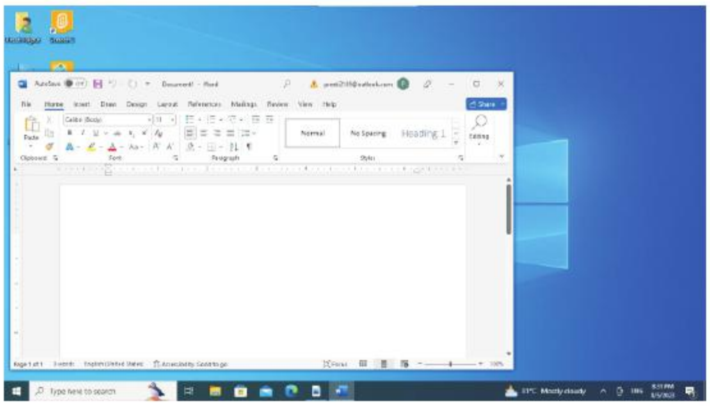
1. Right-click on the tile and select the option Resize.
2. Select the size of your choice.

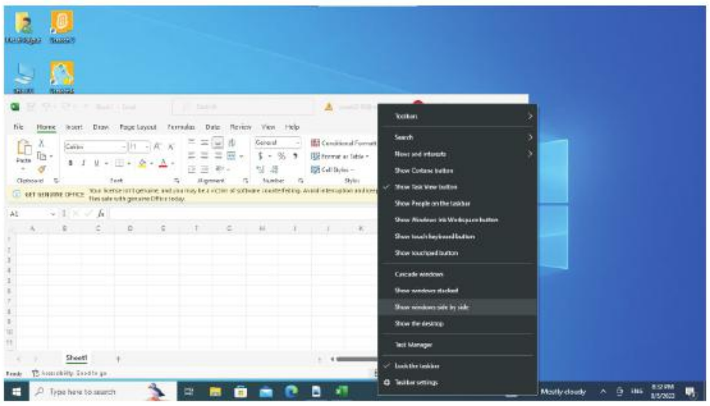


Displaying Multiple Windows at a Time

Follow these steps to open multiple windows at the same time:

1. Open any two applications like Excel 2016 and Word 2016.
2. Right-click on the blank portion of the 'Taskbar'.
3. Click on Show windows side by side option from the menu. You will observe that both the windows will be displayed side by side.





COMPUTER ETIQUETTES

While working on the computer, relax your hands and shoulders.
Take regular breaks, every 5-10 minutes. Do some stretching exercises regularly.
Put up a poster in the computer-area to remind everyone to do stretches.
If you are not typing or using the mouse, relax your hands in your lap.

Let's Implement

A **Tick (✓) the correct answer.**

1. Windows acts as an interface between the user and the ____________.
   a. software ◯ b. hardware ◯
   c. Both a & b ◯ d. None of these ◯
2. The window that appears on the monitor when we switch on a computer is called a ____________.
   a. desktop ◯ b. icons ◯
   c. taskbar ◯ d. notification bar ◯
3. The Start button is present at the ____________ corner of the Taskbar.
   a. bottom left ◯ b. bottom right ◯
   c. top left ◯ d. top right ◯
4. ________________ enables us to connect to and browse the Internet.
   a. This PC ◯ b. Internet Explorer ◯
   c. Both a & b ◯ d. None of these ◯
5. ______________ provides a direct route to a specific application, document or a folder.
   a. Network ◯ b. Shortcut icon ◯
   c. This PC ◯ d. None of these ◯

B **Fill in the blanks.**

1. A long horizontal bar present at the bottom of the desktop is called ______________.
2. The ______________ of the Taskbar shows the programs that are open.
3. The small pictures on a desktop are called ______________.
4. ______________ stores the deleted files and folders till they are permanently deleted.
5. You can pin the ______________ tiles to the start menu.

C **Write True or False.**

1. Desktop appears when we switch on a computer. ☐

2 Small pictures on the desktop are called graphics.

3. Icons are the live tiles.

4. Right-click on the app that you want to add to the 'Start' menu.

5. Live tiles display the latest updates.

D Write the steps to perform the following tasks.

1. Pin tiles to the Start menu.
2. Resize a tile.
3. Open multiple windows at the same time.

E Name the following.

1. Process of loading of operating system in memory ______________
2. GUI operating system ______________
3. Two utilities ______________
4. Any two icons ______________
5. Two Windows 10 apps ______________

F Answer the following questions.

1. Write a note on Taskbar.
2. Discuss the commonly used icons on the desktop.
3. What is the use of This PC?
4. What are live tiles?
5. What should we do to change Date and Time?

Refreshment Corner

- **Draw and colour the logo of Windows 10 in the space given below.** Art Integration

Practical Session

Experiential Learning

1. Create a document in word 2016. Type ten sentences on the topic 'The Plants'. Format your work and save it in D: drive.

   Steps to be Followed

   - Click on Start — Word 2016.
   - Type ten sentences about the topic 'The Planets'.
   - Format the text after typing it.
   - Save your document by the name 'The Plants'.
   - Save it in D: drive.

2. Create a Word 2016 document. Type a paragraph talking about your friends and family. Save it in Documents folder.

   Steps to be Followed

   - Click on Start — Word 2016
   - Type a paragraph about your friends and family.
   - Format the text after typing it.
   - Save your document by the name 'Friends and Family'.
   - Save it in Documents folder.

RACKING YOUR BRAIN

Life Skills and Values

What will you do if you see your friend deleting the files from your computer without your permission?

Just Have a Discussion

Communication

Discuss the different apps of Windows 10 in the class in detail.

Teacher's Note

Explain the use of Windows 10 to the students in detail.

3

Scan Me

ADVANCED FEATURES OF PAINT 3D

Outlines

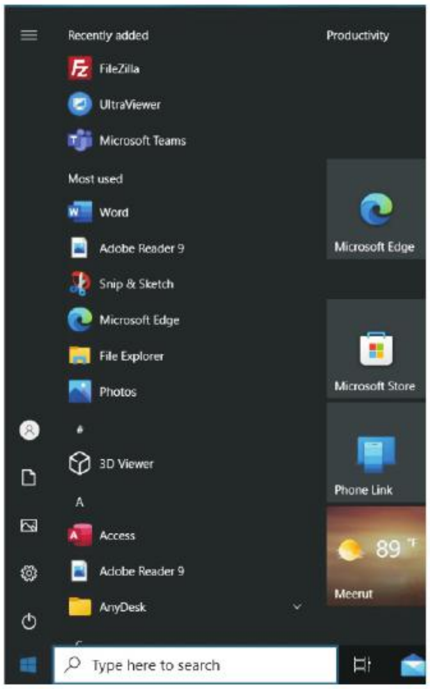
- Starting Paint 3D
- Using 3D Library
- Adding Text
- Adding Effects
- Working with Canvas Tool
- Group/Ungroup objects
- Inserting Stickers

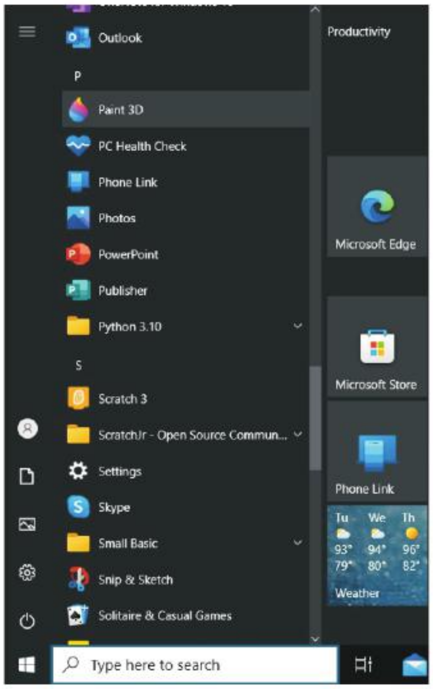
Paint 3D is used to draw shapes like cubes, cuboids, cylinders, cones, spheres and doughnuts. You can rotate and turn these shapes to see different views of the 3D shapes.

Starting Paint 3D

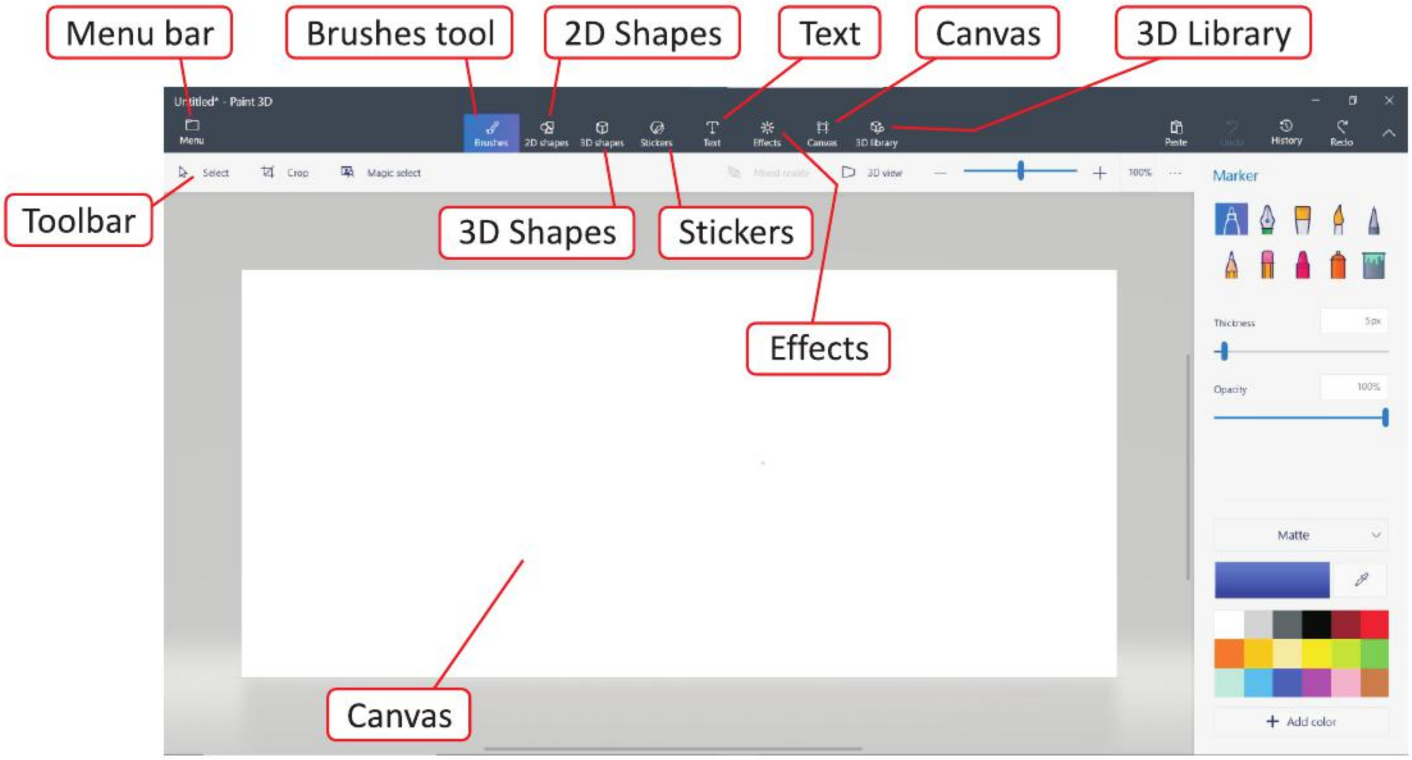
To open Paint 3D, follow the steps given below:

1. Click on the Start button.
2. Click on Paint 3D.
3. The Paint 3D Welcome screen appears. The Welcome screen shows two options: New and Open. Click on the New option to start a new drawing.





You will get the Paint 3D main screen.



Smart Tip

You can also start Paint 3D by typing Paint 3D in the Search box and clicking on Paint 3D application.

Working with Canvas Tool

The Canvas button is used to change the size of the canvas. When you click the Canvas button, the Canvas pane appears on the right showing the following options:

Show Canvas

It shows/hides the Canvas.

Transparent Canvas

It turns the canvas transparent/solid.

Resize Canvas

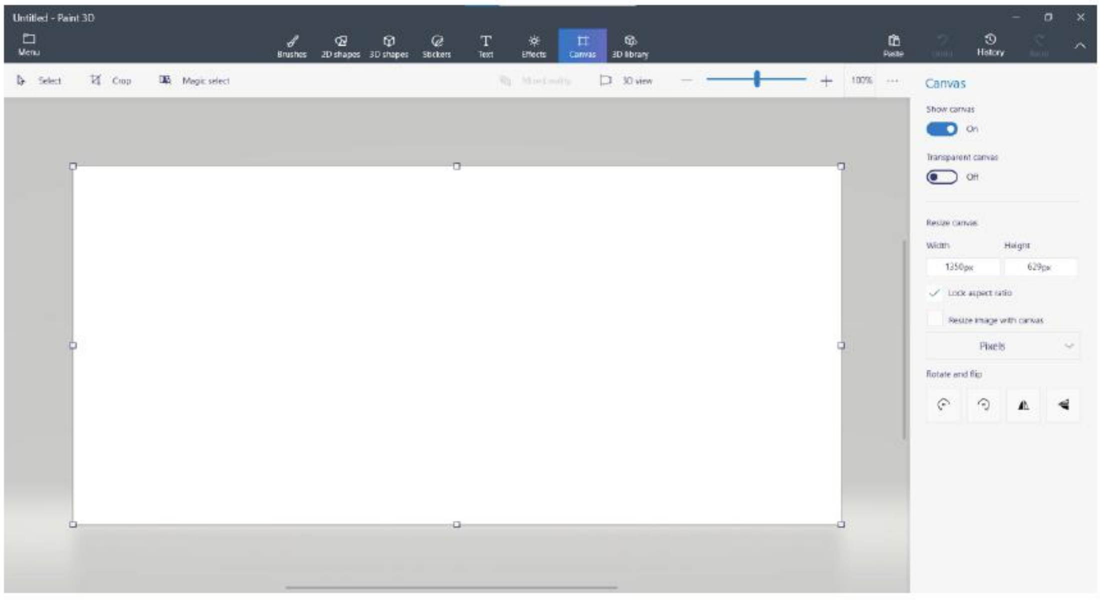
By default, canvas is measured in terms of percentage and is set as 100% long and 100% wide. You can change these values as and when needed. You can also change the unit from percentage to pixel.

Lock Icon

The small lock icon locks/unlocks the aspect ratio. When checked, the two values Width and Height — will always change in the current proportion.

Rotate and Flip

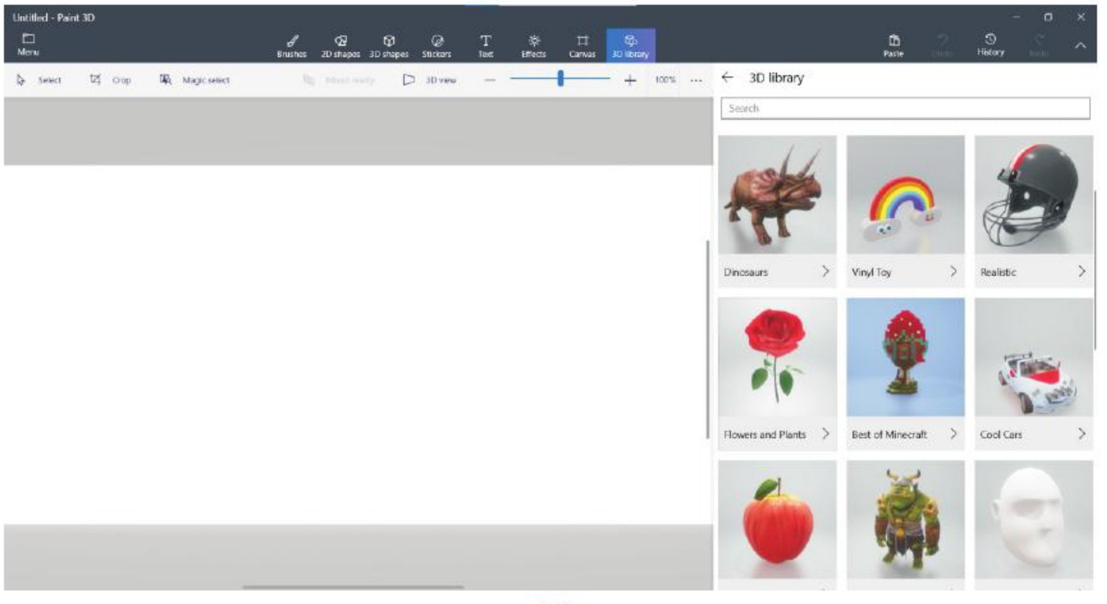
The four buttons available at the bottom, can be used to rotate or flip the canvas.



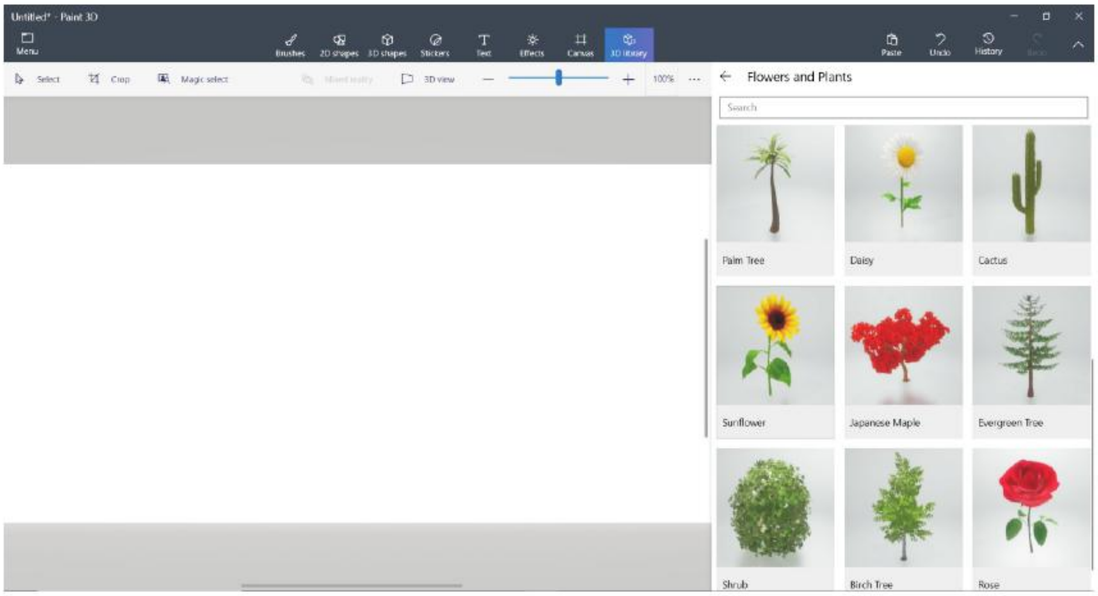
Using 3D Library

You can insert any ready-made 3D objects in your drawing from the 3D library. For this, follow these steps :

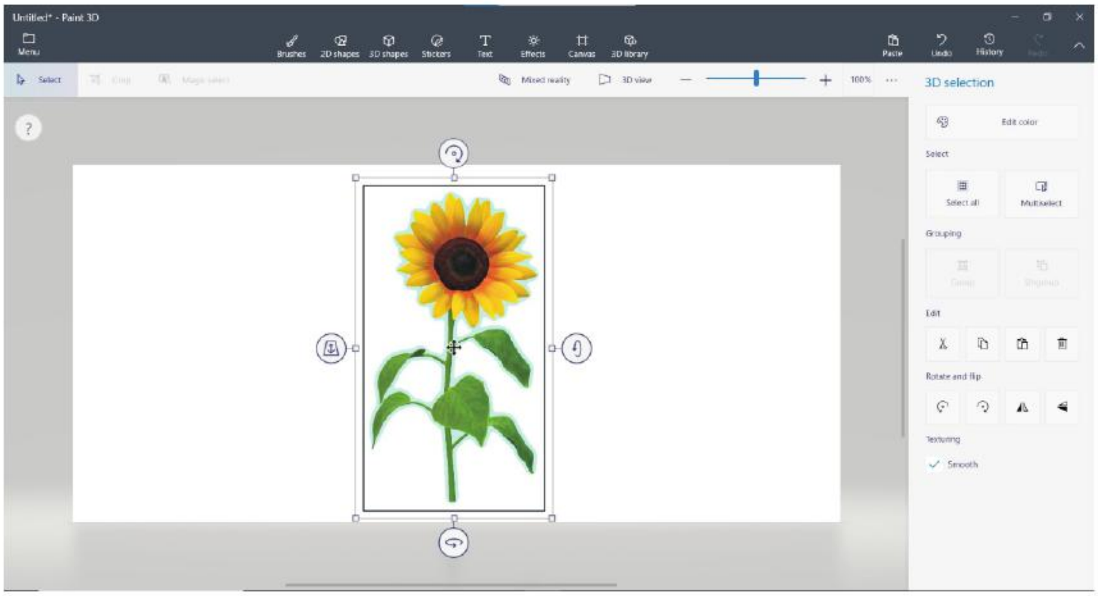
1. Click on the 3D library option.
2. Select a category (for example, Flowers and Plants) by clicking on it.



3. Select a 3D object and click on it to insert.



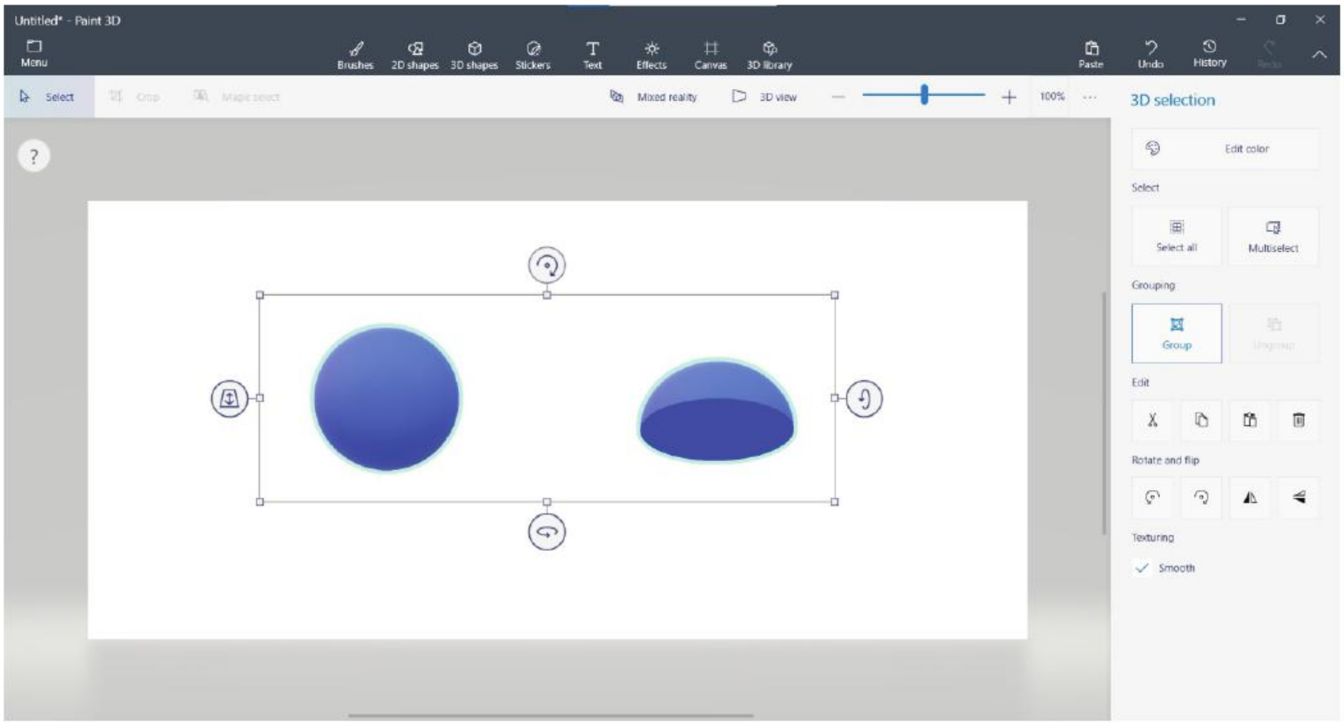
4. Rotate the image using the Rotation handles, if required.



Group/Ungroup Objects

The images (objects) that you draw are individual objects. If you want to treat them as one object, you need to group them. To make a group of objects, follow these steps:

1. Select the objects by clicking them one-by-one while keeping the Shift key pressed.
2. Click Group button in the right pane. The selected objects will become a single object.



To ungroup them, select the Ungroup button.

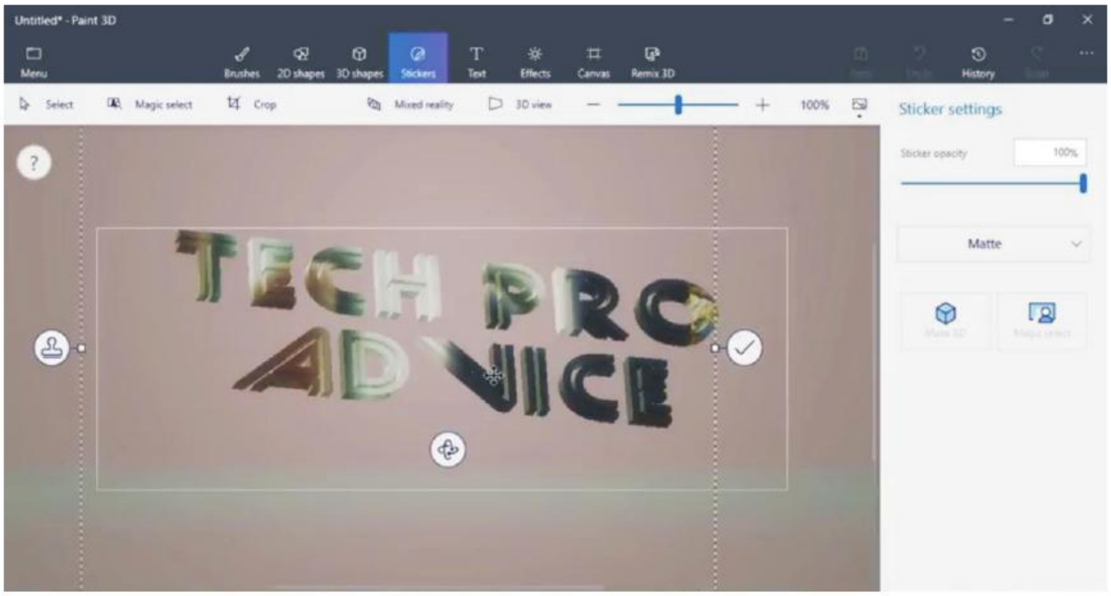
Info Zone

You can also click Select All option in the right pane if you want to select all the objects.

Adding Text

The Text tool is used to write 2D and 3D text. Writing 2D text is straightforward—just click and type. To create 3D text, follow these steps:

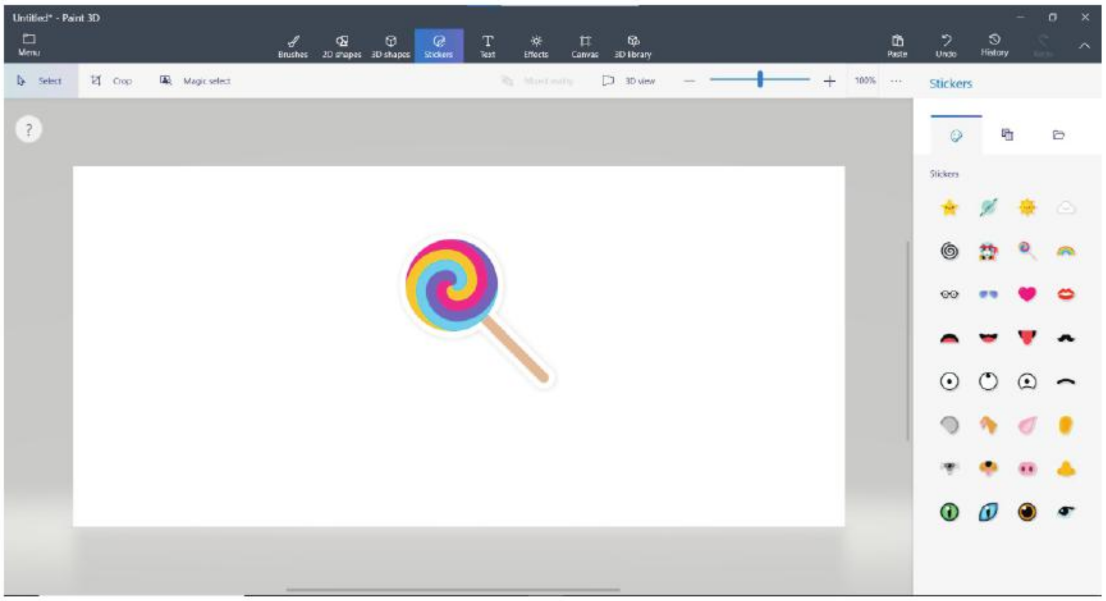
1. Click on the Text tool.
2. Select 3D Text option in the 3D Text pane.
3. A dotted rectangular box with options to rotate the text box will appear with a blinking cursor inside it. Select the font name, font size, font color and set bold, italic, underline, and text alignment (left, center, or right) properties appropriately.
4. Click on the canvas and start typing the text. When you are done, click outside the text box.
5. The controls of 3D text will appear around the text. You can use the Rotate tool to rotate the text and Move back and forth tool to move the text back and forth of the canvas.



Inserting Stickers

The Stickers tool is used to insert stickers into our canvas. To do so, follow these steps:

1. Click on the Stickers tool. The Stickers pane appears on the right with three options tabs — Stickers, Texture, and Custom stickers.
2. Click on the Stickers tab.
3. Select the desired sticker. Now drag and drop it to the canvas.

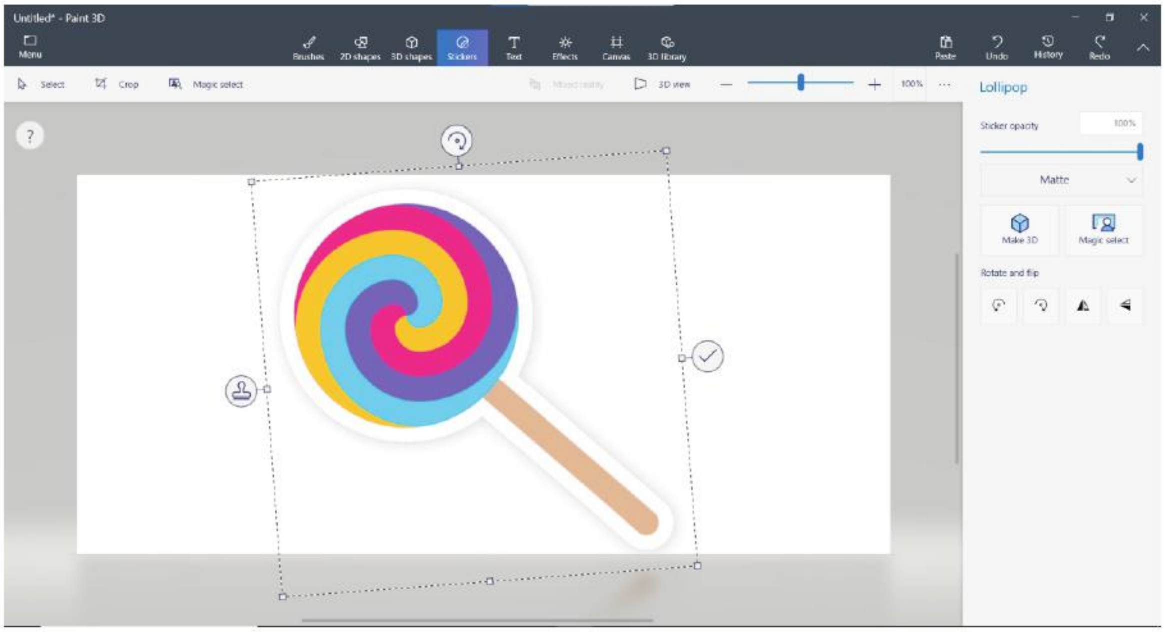


Modifying a Sticker

You can make changes to the sticker that you have inserted. You can change the size, position and appearance of the sticker.

Resizing the Sticker

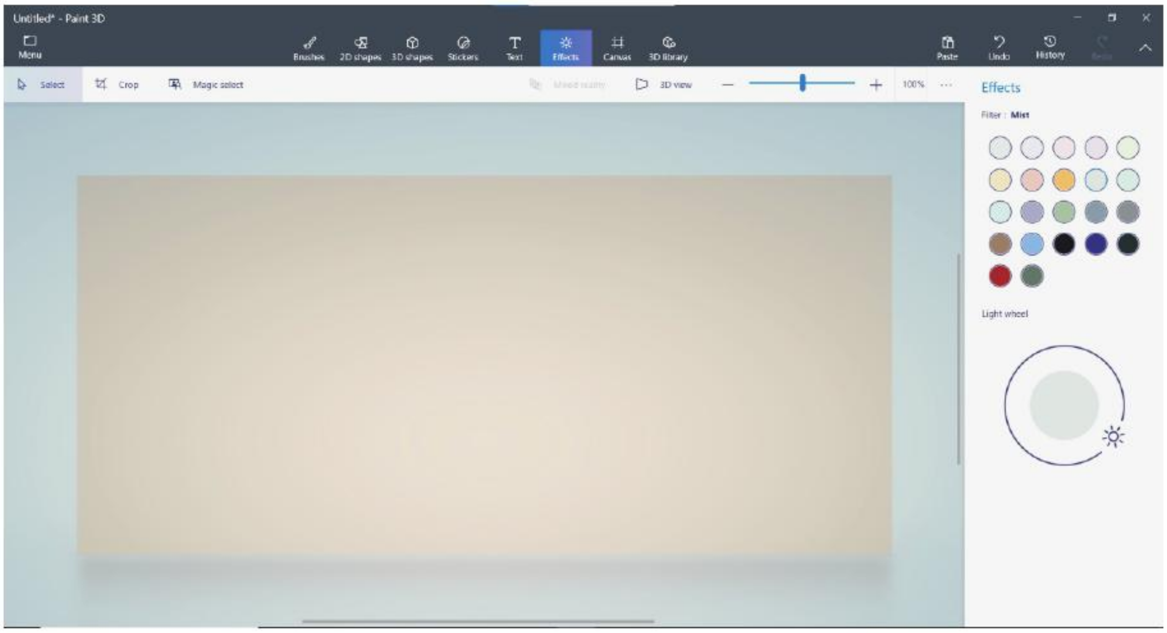
After adding a sticker, you can change its size by dragging the side or corner handles.



Adding Effects

The Effects tool is used to include effects. To add effects, follow these steps:

1. Open your saved Paint 3D file.
2. Click on the Effects tool. The Effects pane will appear on the right-hand side.
3. Select the effect as per your choice.
4. To enhance or enlighten the effect, rotate the icon on the Effects pane.



COMPUTER ETIQUETTES

Ask the teacher every time you want to turn the computer on or off. If a wire has come out, ask your teacher to put it back.

Let's Implement

A **Tick (✓) the correct answer.**

1. The __________ buttons available at the bottom, can be used to rotate or flip the canvas.
   a. Three ◯ b. four ◯ c. five ◯
2. The __________________ button is used to change the size of the canvas.
   a. Stickers ◯ b. Text ◯ c. Canvas ◯
3. You can select the objects by clicking them one-by-one while keeping the ____________ key pressed.
   a. Ctrl ◯ b. Alt ◯ c. Shift ◯
4. The Stickers pane consists of __________________ options tabs.
   a. Two ◯ b. three ◯ c. four ◯

B **Fill in the blanks.**

1. The welcome screen shows two options: __________________ and __________________.
2. By default, canvas is measured in terms of __________________.
3. The __________________ tool is used to insert stickers into our canvas.
4. The __________________ tool is used to include effects.
5. To ungroup objects, select the __________________ button.

C **Write True or False.**

1. Effects tool is used to add different stickers. ☐
2. When you click the Canvas button, the Canvas pane appears on the right. ☐
3. Transparent canvas turns the canvas transparent/solid. ☐
4. You cannot change unit from percentage to pixel. ☐
5. The Text tool is used to write 2D text only. ☐

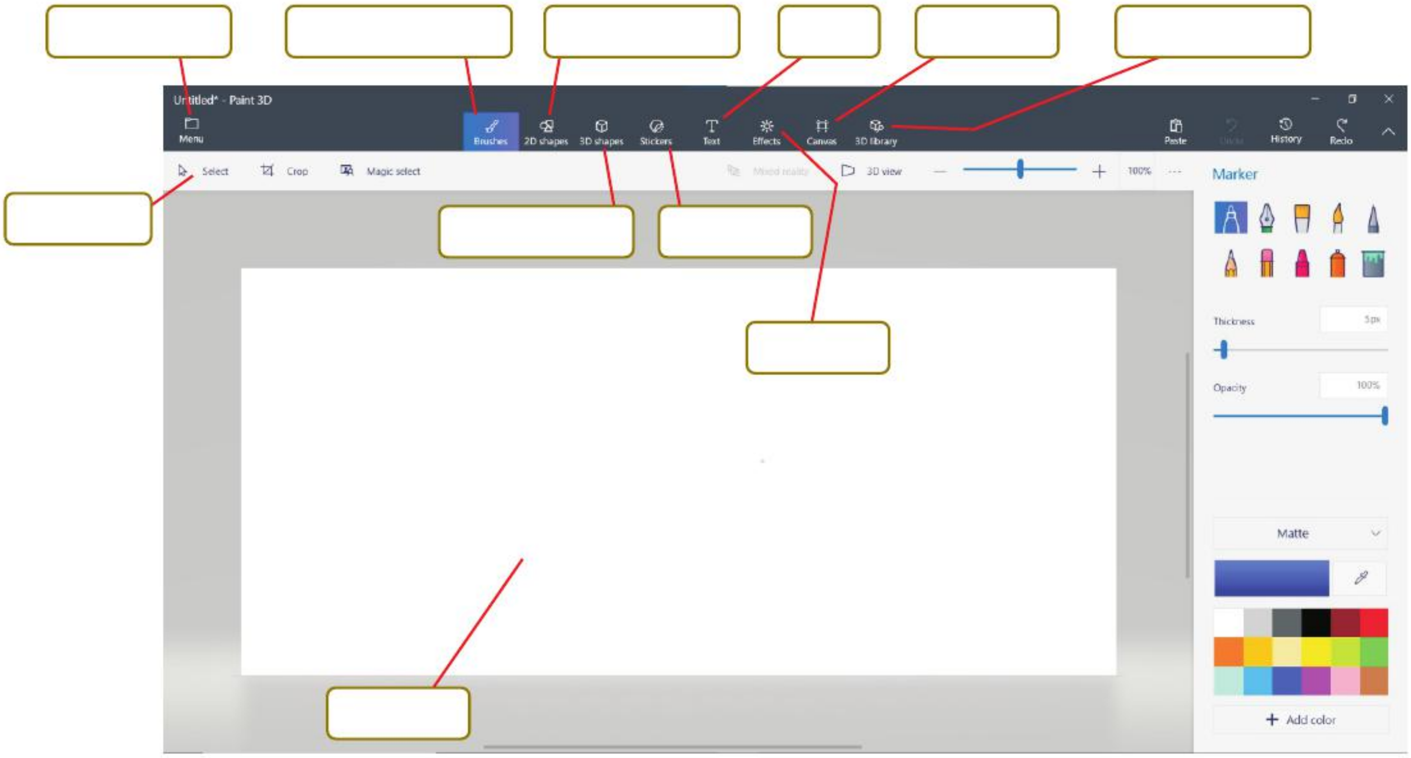
D **Answer the following questions.**

1. What is Paint 3D?
2. Write a note on Canvas tool.

3. What is the use of 3D library?
4. Can we modify or resize a sticker?
5. What is Effects tool?

Refreshment Corner

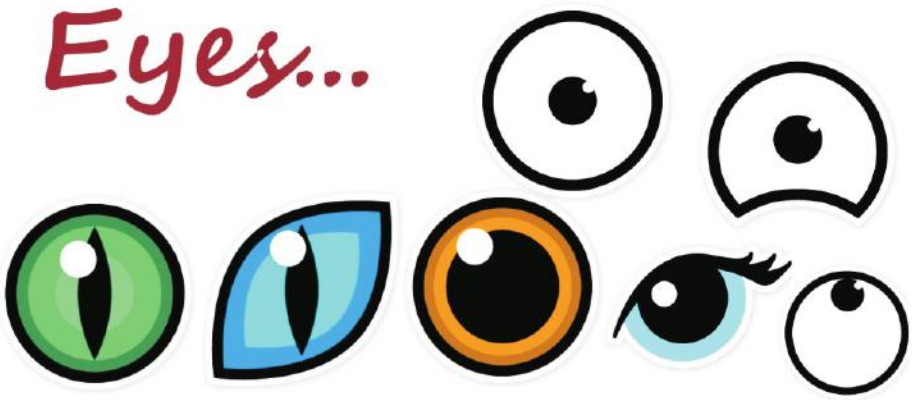
Label the figure given below.



Practical Session

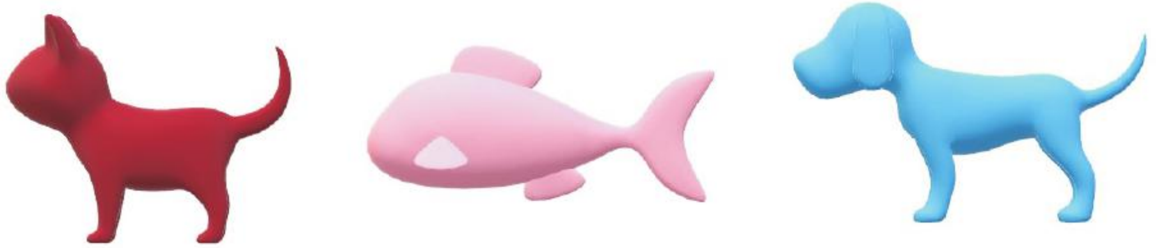
Experiential Learning

A Draw the following drawing using Text tool and Stickers in Paint 3D.



B **Draw the following drawing using Text tool and 3D Library in Paint 3D.**

C **Draw the following pictures and group them.**



RACKING YOUR BRAIN

Life Skills and Values

Your friend is unable to insert stickers into the canvas. What will you do? Justify your answer.

Just Have a Discussion

Communication

Discuss the different components of Paint 3D window in the class in detail.

Teacher's Note

- Explain the difference between 2D and 3D text.
- Demonstrate the use of Canvas tool to the students.

4

MORE ON WORD 2016

Outlines

- Applying Drop Cap Effect
- Inserting WordArt
- Adding a Text Box
- Inserting Shapes

MS Word 2016 allows you to insert more than just text in a document. You can insert shapes, pictures, icons, etc. They are together known as objects. You can insert these objects in your document to make it visually appealing. Let us take them up one by one.

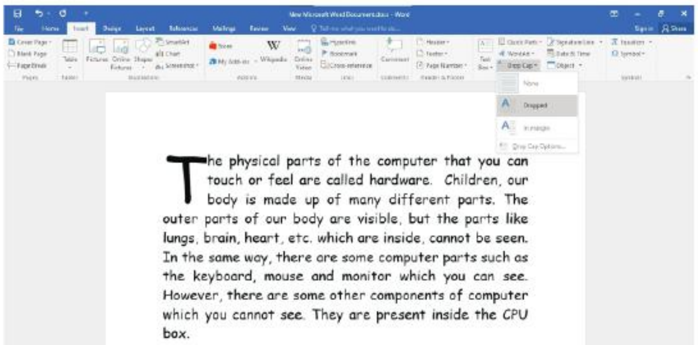
Applying Drop Cap Effect

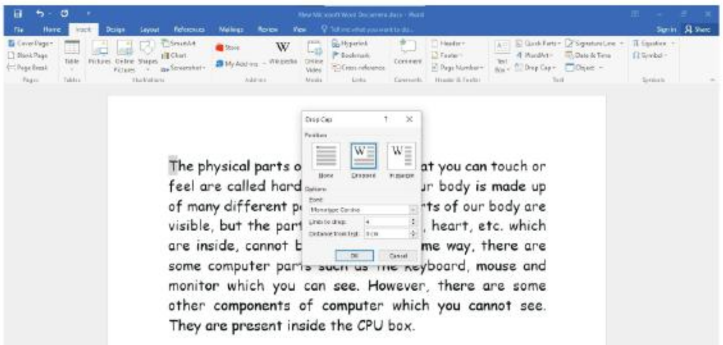
Drop Cap is a text formatting option in Word that allows a user to enlarge only the first letter of a paragraph. To apply the drop cap effect, follow these steps:

1. Select the first word of your document.
2. Click on the Insert tab.
3. Select the Drop Cap option present in the Text group.
4. Select the desired option from the drop-down menu.

Or

a. Select the Drop Cap option from the list.





b. The drop Cap dialog box appears. Select the Dropped option.

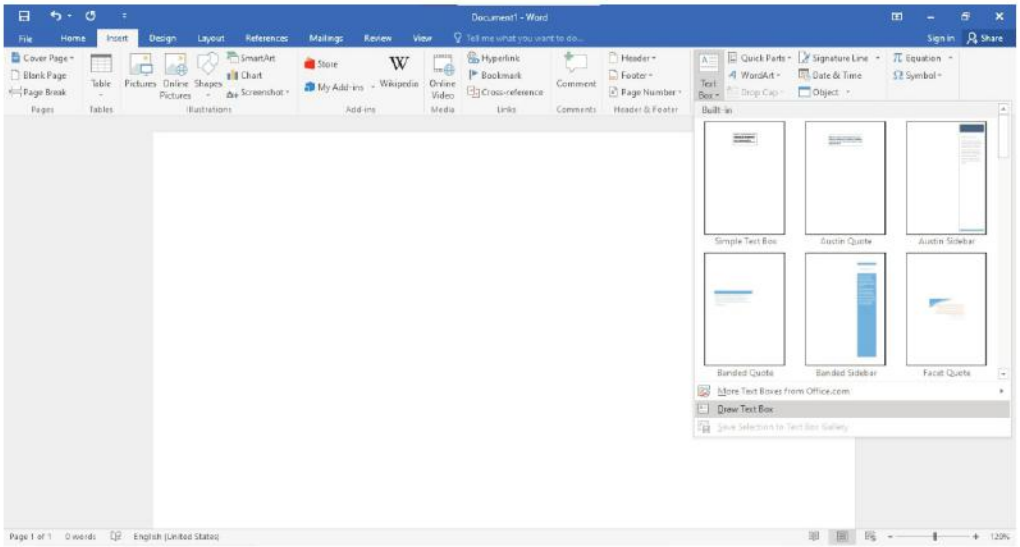
c. Select the options like Font, Lines to drop, if required.

Click on the OK button and observe the change.

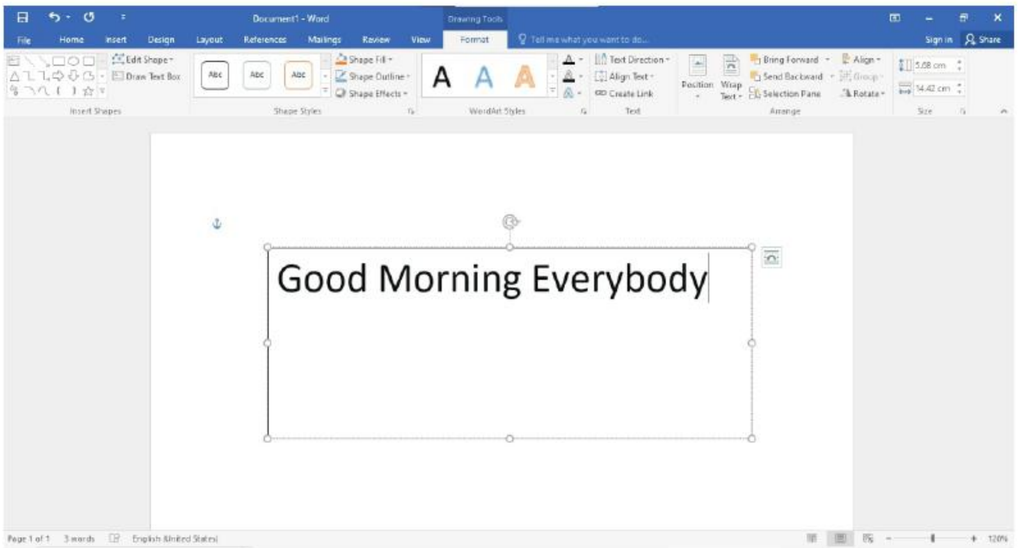
Adding a Text Box

A text box is a placeholder used to enter the text. To create a text box, follow these steps:

1. Click on the Insert tab.
2. Click on the Text Box option in the Text group.
3. Now, select the Draw Text Box option from the drop-down menu.



4. The pointer changes to + . Click and drag the mouse pointer on the document to create a text box.



The text box will be drawn with the cursor inside it. You will observe that the Format tab appears automatically on the ribbon.

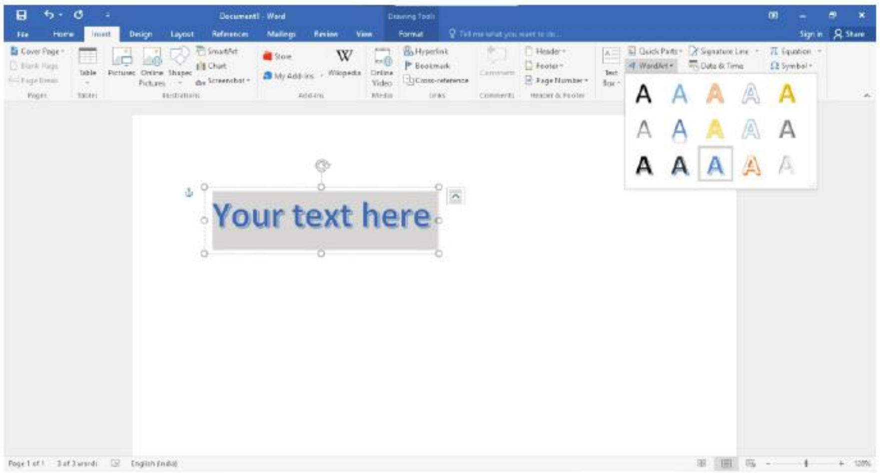
5. Click anywhere outside the text box to return to your document.

Info Zone

You can change the style of text box by selecting the text box and clicking on the More drop-down arrow in the Shape Styles group on the Format tab. Move the pointer over the styles and select the desired style.

Inserting WordArt

WordArt is a text-styling feature. The WordArt gallery includes different styles that can be applied to any text. The shape and formatting characteristics of a WordArt object, are called WordArt styles. To insert a WordArt, follow these steps:



1. Place the cursor at the point where you want to insert WordArt.
2. Click on the Insert tab.
3. Click on the WordArt option in the Text group. The WordArt gallery appears.
4. Select the desired WordArt style. A placeholder appears in the document.
5. Replace the placeholder with the text you want.

When you click on the inserted WordArt, the Format tab appears. It has many options for making changes to the WordArt.

The use of some options on the Format tab is discussed in the table given below:

Option	Uses
Text Direction	This option in the Text group lets you change the direction of the WordArt text.
Edit Shape	This option in the Insert Shapes group lets you change the shape.
Text Fill	This option in the WordArt Styles group lets you fill different colours and gradient in the WordArt text.
Text Effects	This option in the WordArt Styles lets you add various effects like Shadow, Reflection, Glow, Bevel, 3-D Rotation and Transformation to the text.

Inserting Shapes

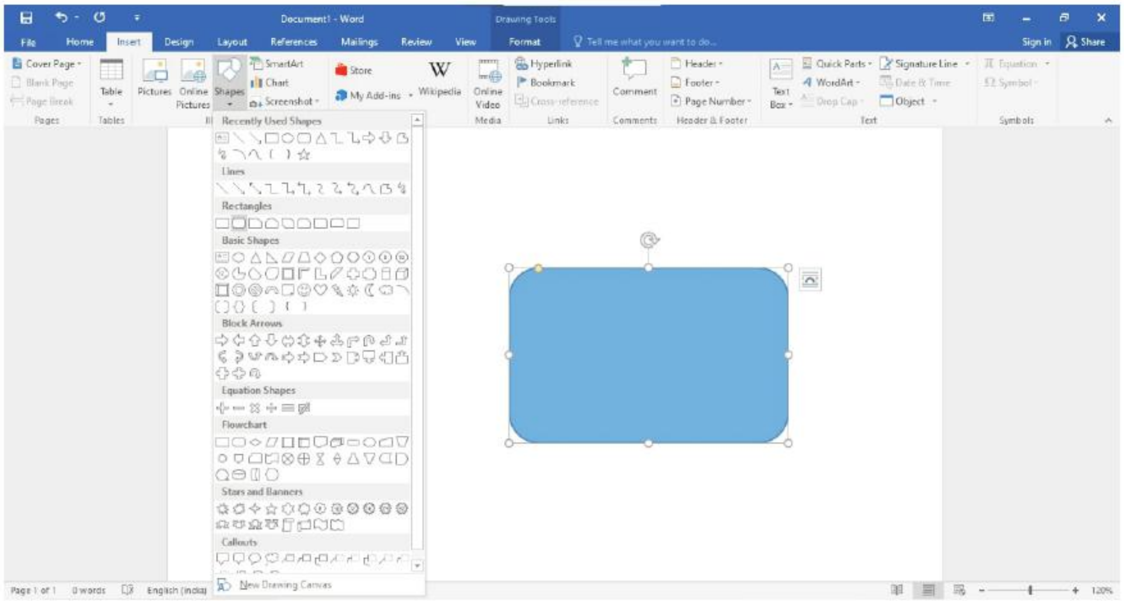
You can insert a variety of shapes like lines, rectangles, stars, etc. This feature is very useful as it lets you improve the presentation of a document by including diagrams.

The shapes in Word are divided into eight main categories:

- Lines
- Rectangles
- Basic Shapes
- Block Arrows
- Equation Shapes
- Flowchart
- Stars and Banners
- Callouts

To insert any shape in Word, follow these steps:

1. Click on the Insert tab.

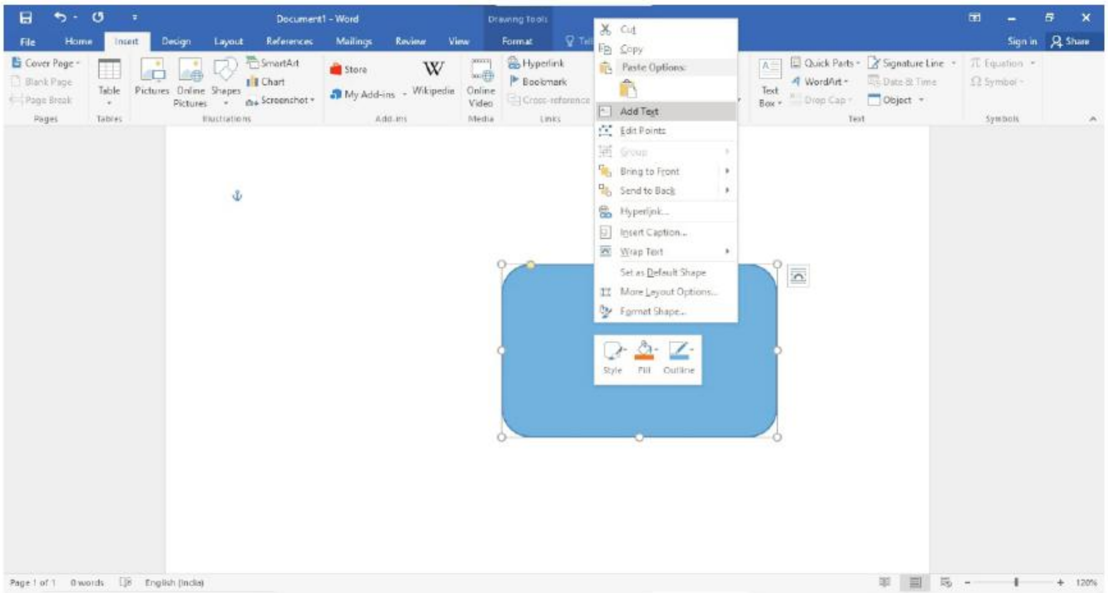


2. Click on the down arrow under Shapes option in the Illustrations group.
3. Select the desired shape in the gallery that opens.

4. The cursor changes to a + sign. Click, hold and drag the mouse to draw the shape.

Adding Text in Shapes

You can also insert text inside the shape. To insert text inside the shape, follow these steps:



1. Right-click the shape in which you want to insert text.
2. Select Add Text option.
3. Start typing the text.

When you click on the inserted shape, the Format tab appears.

The use of some options on the Format tab is described in the table given below.

Option	Uses
Shape Styles	The Shape Styles option lets you apply preset fill colours and outline colours to the selected shape. To display the Shape Styles gallery, click on the More button.
Shape Fill	This option is used to fill the selected shape with a solid colour.
Shape Outline	This option is used to specify the colour, thickness and line style for the outline of the selected shape.
Shape Effects	This option is used to apply shape effects like Shadow, Reflection, Glow, Soft Edges and 3-D Rotation to the selected shapes.

Let's Implement

A Tick (✓) the correct answer.

1. ____________ is a text-styling feature in Word 2016.
   a. WordArt ◯ b. Text box ◯ c. Shapes ◯
2. ____________ option is not present in the Text group under the Insert tab.
   a. Text box ◯ b. WordArt ◯ c. Shapes ◯
3. A ____________ is a placeholder used to enter text.
   a. Text box ◯ b. WordArt ◯ c. Picture ◯
4. The shapes in Word are divided into ____________ main categories.
   a. six ◯ b. seven ◯ c. eight ◯
5. ______________ option is used to fill the selected shape with a solid colour.
   a. Shape Outline ◯ b. Shape Fill ◯ c. Shape Effects ◯

B Fill in the blanks.

1. ____________ is a text formatting option that allows a user to enlarge only the first letter of a paragraph.
2. The ____________ gallery includes different styles that can be applied to any text.
3. The ____________ option in the WordArt Styles group lets you fill different colours and gradient in the WordArt text.
4. Shapes option is in the ____________ group.
5. Right-click the shape in which you want to insert ____________.

C Write **True** or **False**.

1. WordArt is a text-styling feature. ☐
2. When you click on the inserted WordArt, the Format tab appears. ☐
3. The Shapes option is present on the Home tab. ☐
4. The shape and formatting characteristics of a WordArt object, are called WordArt styles. ☐
5. You cannot insert text inside a shape. ☐

D Answer the following questions.

1. What is the use of Drop Cap option?
2. What is WordArt?
3. What is the use of Text box? Can we change its style?
4. How will you insert any shape in a document?
5. What is the use of Shape Effects option?

Refreshment Corner

A **Identify the following options and write their names.**

1. __________ 2. __________ 3. __________ 4. __________ 5. __________

Practical Session

Experiential Learning

1. Prepare a poster on 'Save Water'. Use your imagination and design an attractive poster.
2. Draw the figures given below using Shapes in MS Word 2016.



RACKING YOUR BRAIN

Life Skills and Values

How will you help your friend who is unable to insert shapes in a document after trying several number of times?

Just Have a Discussion

Communication

Discuss how you can you apply WordArt in a document in the class.

Teacher's Note

- Demonstrate to the students how the appearance of a Word document can be improved.
- The teacher should demonstrate to the students how to insert objects in MS Word 2016.

QUICK REVIEW 1

Tick (✓) the correct option.

1. First generation computers:
   a. used vacuum tubes b. used integrated circuits
   c. were built after 1955 d. All of these
2. The Start button is present at the ____________ corner of the Taskbar.
   a. bottom left b. bottom right
   c. top left d. top right
3. You can select the objects by clicking them one-by-one while keeping the ____________ key pressed.
   a. Ctrl b. Alt
   c. Shift d. Tab
4. The shapes in Word are divided into ____________ main categories.
   a. six b. seven c. eight d. nine
5. Second generation computers used:
   a. vacuum tubes instead of transistors
   b. transistors instead of vacuum tubes
   c. integrated circuits
   d. None of the above
6. ________________ enables us to connect to and browse the Internet.
   a. This PC b. Internet Explorer
   c. Both a & b d. None of these
7. The Stickers pane consists of ________________ options tabs.
   a. Two b. three
   c. four d. five
8. ____________ is a text-styling feature in Word 2016.
   a. WordArt b. Text box
   c. Shapes d. None of these
9. ______________ provides a direct route to a specific application, document or a folder.
   a. Network b. Shortcut icon
   c. This PC d. None of these
10. The _________ buttons available at the bottom, can be used to rotate or flip the canvas.
    a. three b. four
    c. five d. six

5

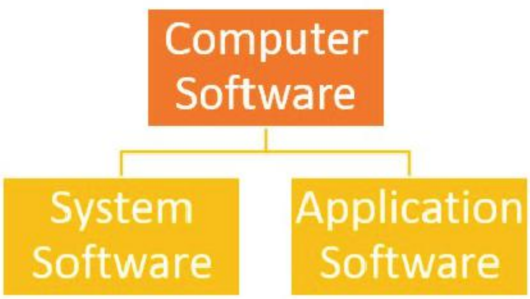
MORE ON POWERPOINT 2016

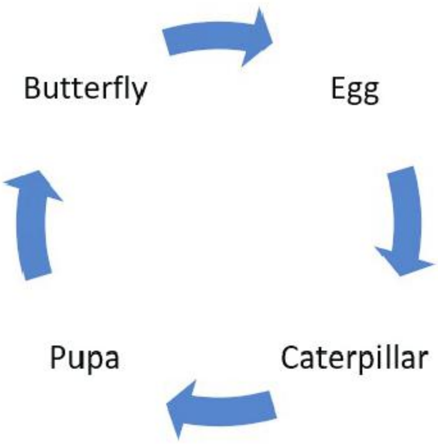
Outlines

- Inserting SmartArt Graphics
- What are Animations?
- Inserting Sound Files
- Inserting a Table
- Applying Animation Effects

Inserting SmartArt Graphics

SmartArt Graphics let you present data or information visually in the form of flowcharts, diagrams, etc. These graphics are a wonderful way to convey your message easily and effectively.

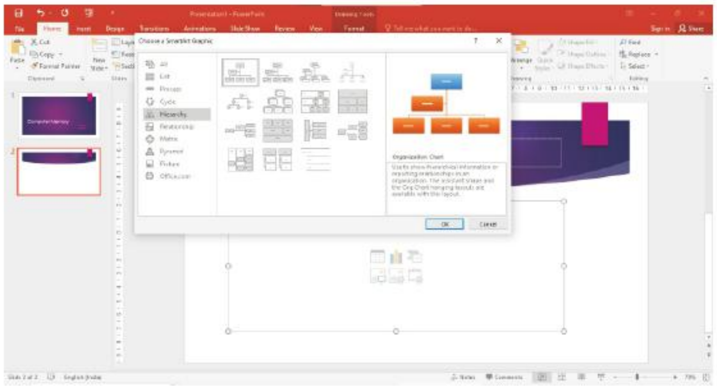


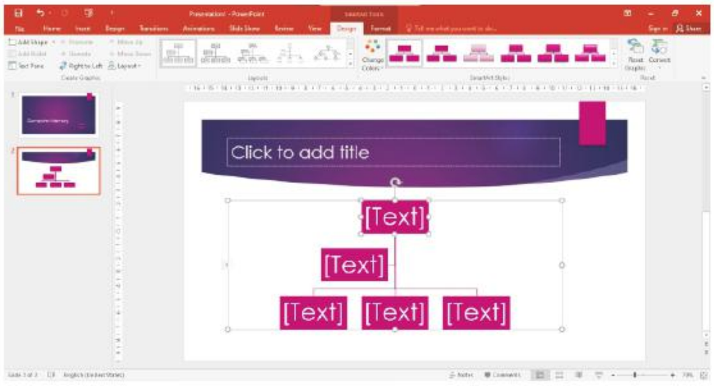


To insert SmartArt graphics, follow these steps:

1. Click on the Title and Content layout.
2. Click on the Insert a SmartArt Graphic icon on the slide. The Choose a SmartArt Graphic dialog box appears.
3. Select a category on the left.

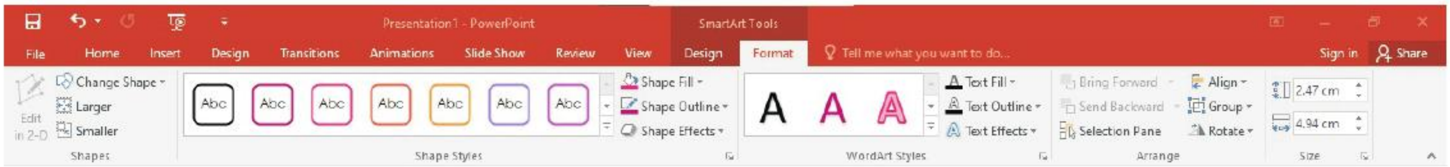
4. Choose the desired SmartArt graphic.
5. Click on the OK button.





The Design and Format tabs appear under SmartArt Tools whenever you click on the inserted SmartArt graphic.

Design tab



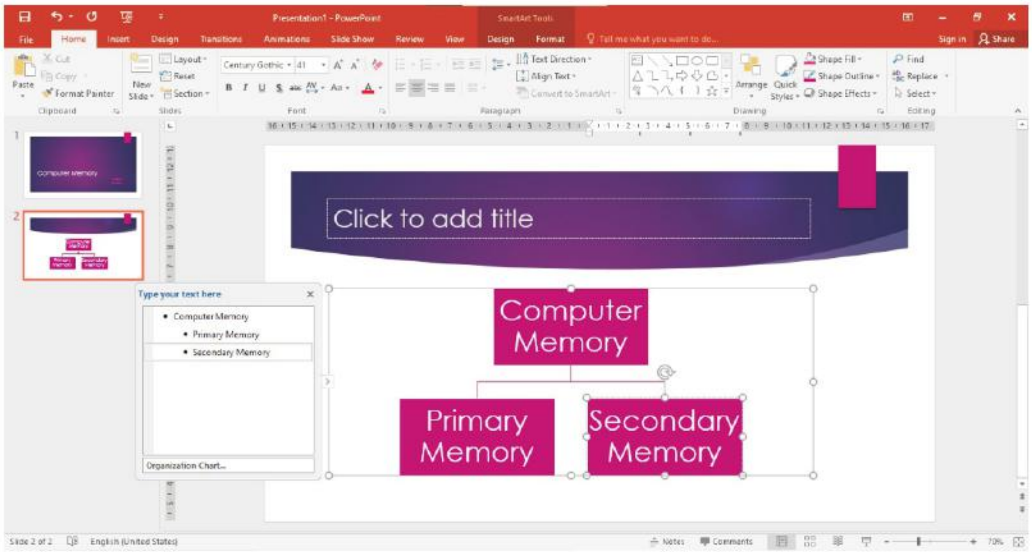
Format tab

To add text to a SmartArt graphic

To type text, do one of the following:

a. Click inside a box in the SmartArt Graphic to add text to it.

b. Click on the [Text] box in the Text pane to add text to it.



Info Zone

Click on the Control present on the left side of the inserted SmartArt, if the Text pane is not visible.

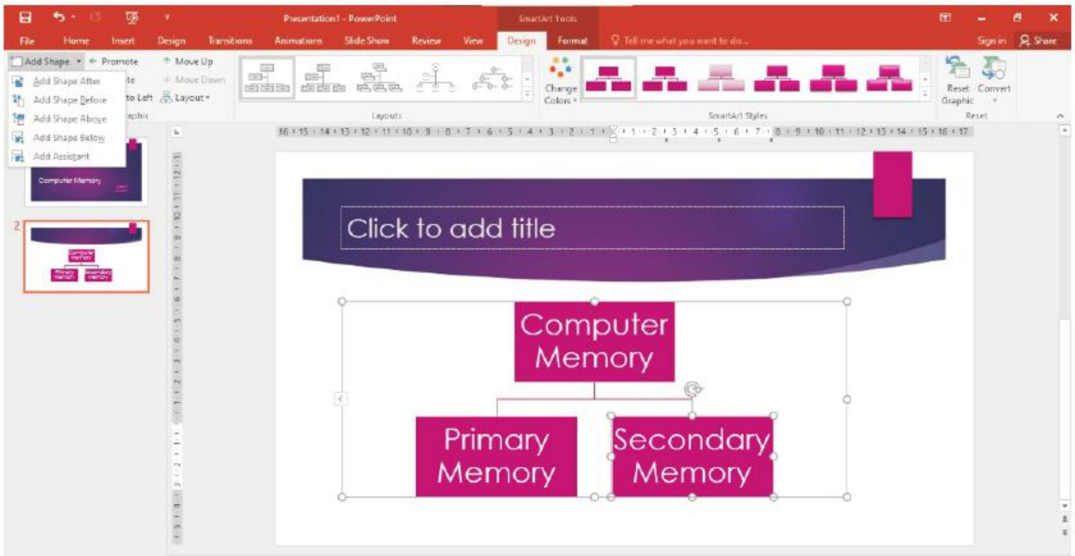
To Add a Shape

You can also add another box under or next to an existing box. For this, click on the arrow next to the Add Shape option in the Create Graphic group on the Design tab under SmartArt Tools. A drop-down list is displayed. Then, select the desired option from the list.

a. To insert a box at the same level:

(i) Choose the Add Shape After option to add a box following the selected box.

(ii) Choose the Add Shape Before option to add a box before the selected box.



b. To insert a box above or below one level:

(i) Choose the Add Shape Above option to add a box above the selected box.

(ii) Choose the Add Shape Below option to add a box below the selected box.

c. To insert an assistant box:

To insert an assistant box, choose the Add Assistant option. This option places the new box below the selected shape with an elbow connecter.

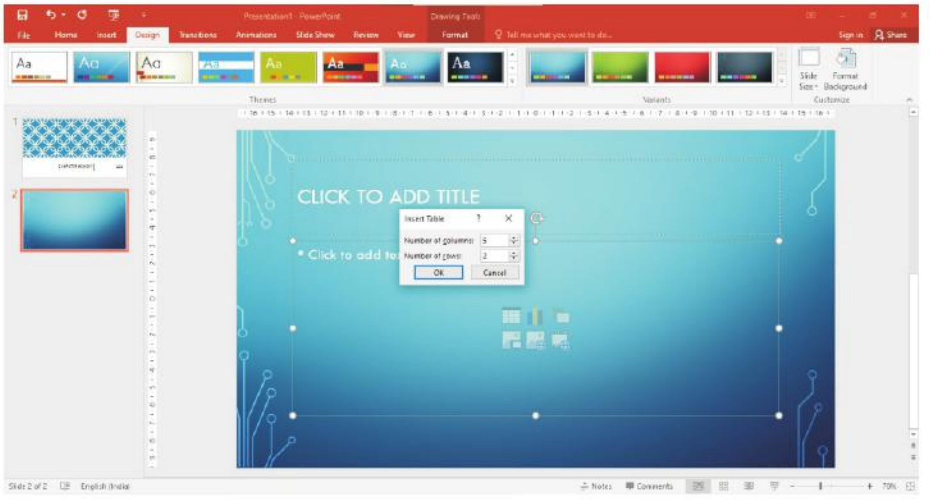
Inserting a Table

Tables are useful for various tasks such as presenting text information and numerical data. A table is a set of data (text and/or numbers) arranged in rows and columns. The intersection of rows and columns forms rectangular boxes called cells.

The vertical series of cells in a table is called a column. The horizontal series of cells in a table is called a row.

To insert a table, follow these steps:

1. Select the Title and Content layout slide in a presentation.
2. Click on the Insert Table icon present on the slide. The Insert Table dialog box appears.
3. Specify the number of rows and columns.
4. Click on the OK button.



What are Animations?

Animations are special effects that you can add to text and objects on a slide. Animations can make the audience pay attention to important points, control the flow of information and make the presentation interesting.

MS PowerPoint offers many animation effects. These have been organised under four categories:

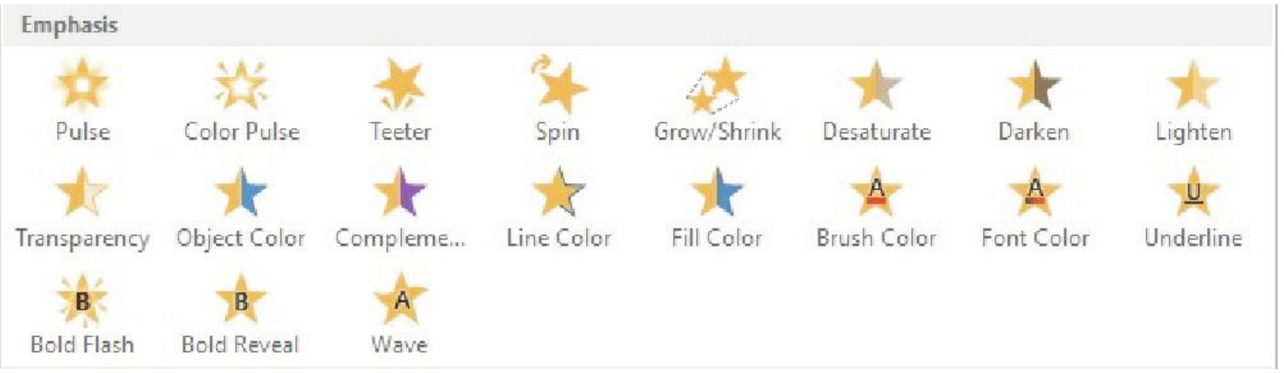
- Entrance Effects
- Emphasis Effects
- Exit Effects
- Motion Paths

Entrance Effects

Entrance effects control the manner in which an object is introduced into the slide during the slide show. Some of these effects are Appear, Fade, Fly In, Bounce, Zoom, etc.

Emphasis Effects

Emphasis effects occur when the object is on the slide during the slide show. They are used to emphasise the object. Some of these effects are Pulse, Spin, Underline, Darken, Wave, etc.

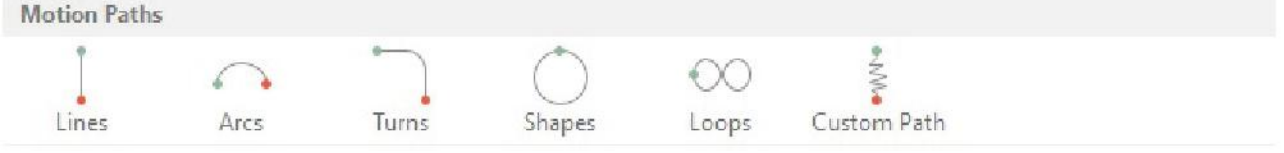


Exit Effects

Exit effects control the manner in which the object exits during the slide show. Some of these effects are Disappear, Fly Out, Split, Wheel, Swivel, etc.

Motion Paths

Motion paths effects enable objects to move from one position to another on the specified path during the slide show. Some of these effects are Lines, Arcs, Turns, Shapes, Loops, etc.



Applying Animation Effects

To add animation effects to text or objects, follow these steps:

1. Select the text or an object that you want to animate.
2. Click on the Animations tab.
3. Click on the More arrow button in the Animation group. The gallery of Animation Effects appears.

4. Click on the desired animation effect.

After you have applied an animation to an object or a text, you will notice that a numbered tag appears near the object on the slide pane. This implies that an animation effect has been applied on the object.

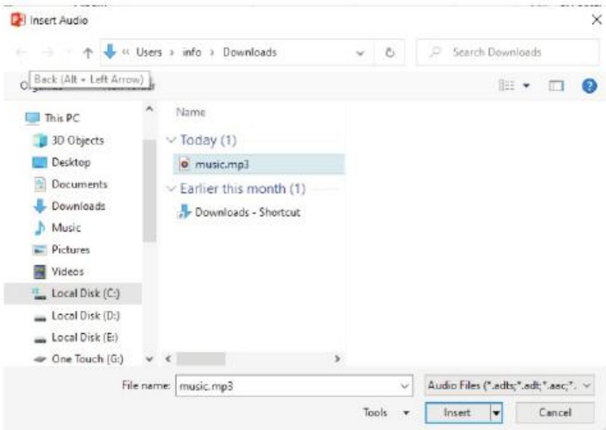
Inserting Sound Files

PowerPoint lets you easily insert songs, voice recordings, etc. in your presentation.

Insert a Sound from an Audio File

To insert a sound file from an audio file in a slide, follow these steps:

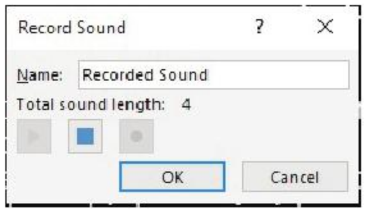
1. Click on the slide on which you want to add an audio file.
2. Click on the Insert tab.
3. Click on the down arrow under Audio option in the Media group. A menu opens.
4. Click on the Audio on My PC... option. The Insert Audio dialog box appears.
5. Select the desired audio file.
6. Click on the Insert button.

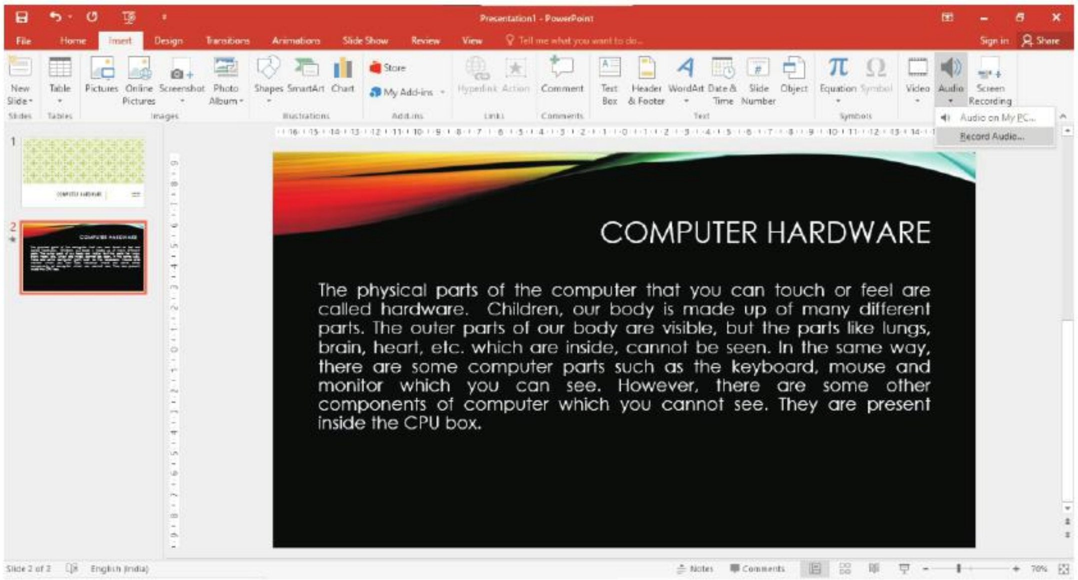


Insert a Recorded sound

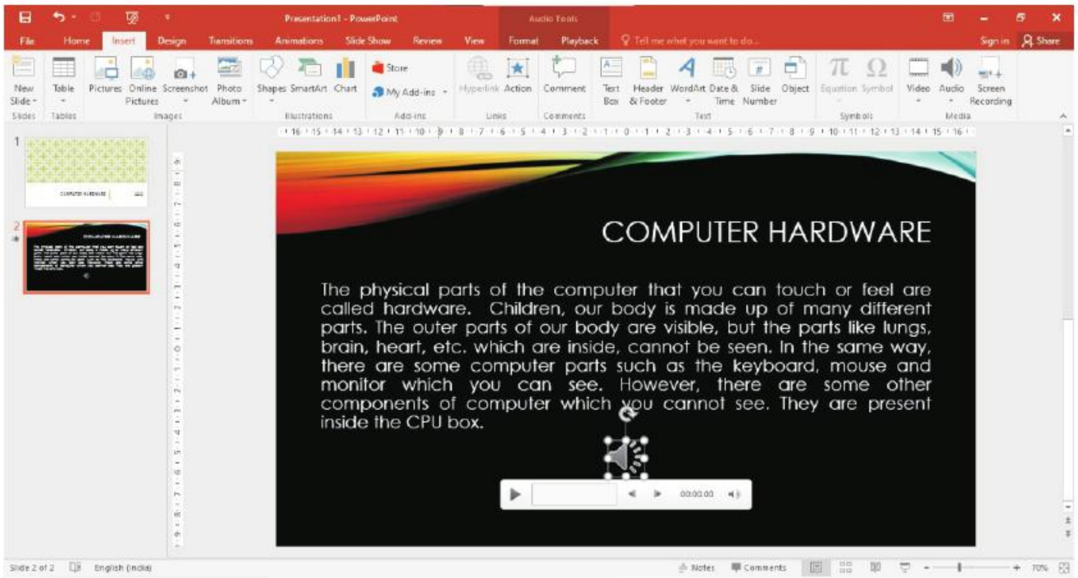
To insert a recorded sound in a slide, follow these steps:

1. Click on the slide on which you want to add an audio file.
2. Click on the Insert tab.
3. Click on the down arrow under Audio option in the Media group. A menu opens.
4. Click on the Record Audio option. The Record Sound dialog box appears.
5. Type a name for the audio recording.
6. Click on the Record button to start recording.
7. Click on the Stop button when you finish recording.
8. You can click on the Play button to listen to the recording.
9. Click on the OK button.





The audio file gets inserted in the slide. Once you have inserted an audio, an audio clip icon appears on the slide.



Let's Implement

A Tick (✓) the correct answer.

1. _______________ is not a category under the Animation effects.
   a. Entrance effects ◯ b. Enlivening effects ◯
   c. Both a and b ◯ d. None of these ◯
2. _______________ effects enable objects to move from one position to another on the specified path during the slide show?
   a. Motion paths ◯ b. Emphasis ◯
   c. Both a and b ◯ d. None of these ◯
3. _________ effects control the manner in which the object exists during the slide show.
   a. Entrance ◯ b. Exit ◯
   c. Both a and b ◯ d. None of these ◯
4. The Audio option is present in the _______________ group on the Insert tab.
   a. Media ◯ b. Create Graphic ◯
   c. Both a and b ◯ d. None of these ◯
5. There are _______________ categories of Animation Effects.
   a. four ◯ b. five ◯
   c. three ◯ d. None of these ◯

B Fill in the blank.

1. ____________ effects control the manner in which the object is introduced into the slide during the slide show.
2. ____________ are special effects that you can add to text and objects on a slide.
3. A vertical series of cells in a table is called a ____________.
4. ____________ let you present data or information visually in the form of flowcharts, diagrams, etc.
5. ____________ effects occur when the object is on the slide during the slide show.

C Write **True** or **False**.

1. SmartArt Graphics are a wonderful way to convey your message easily and effectively. ☐
2. Add Shape option is in the Media group. ☐
3. To insert a table, we should select the Title and Content layout slide. ☐
4. Fly In is an Entrance effect. ☐
5. Wheel is an Emphasis effect. ☐

D **Answer the following questions.**

1. What do you mean by SmartArt Graphics?
2. How can we insert SmartArt Graphics?
3. What are animations?
4. What are the four categories of animation effects? Explain.
5. List out the steps to insert sound from an audio file in a slide.

Refreshment Corner

- **Make a presentation of your class students numbers and take printout of them.**

Practical Session

Experiential Learning

1. Make a presentation on 'Dance forms of India'. Try to make the presentation informative and interesting by making use of smart graphics animation effects and sound.
2. Prepare a presentation on 'Computer Software'. Make use of SmartArt graphics, animation effects and sound.

RACKING YOUR BRAIN

Life Skills and Values

Your friend starts a slide show of his/her presentation. But, he/she is not able to exit the slide show then what will you do? Justify your answer.

Just Have a Discussion

Communication

Divide the class into two groups and have a discussion on— 'Various Animations Effects'.

Teacher's Note

The teacher should demonstrate animation effects to the students. The students should be told that animations make a presentation interesting but they should not be overused. Animations should be used wisely to emphasize important points.

6

INTRODUCTION TO EXCEL 2016

Scan Me

Outlines

- What is Microsoft Excel?
- Data Types
- Moving Around in a Worksheet
- Auto Fill
- Working with Worksheets
- Starting Microsoft Excel
- Entering Data in a Worksheet
- Creating a New Workbook
- Performing Calculations
- Saving a Workbook

What is Microsoft Excel?

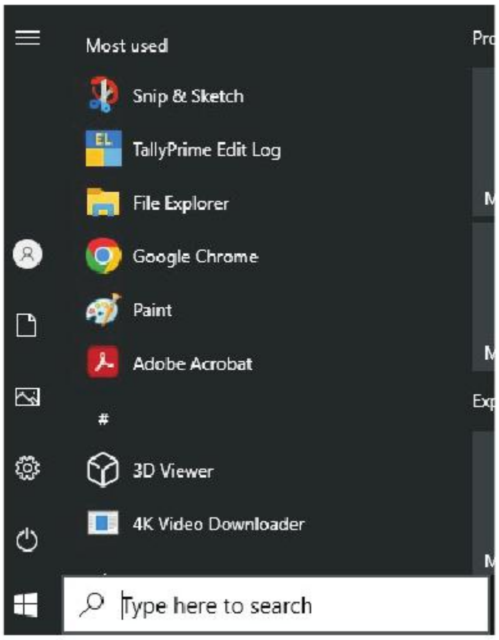
Excel 2016 is a software that is a part of the Microsoft Office suite. This software is mainly used to create tables with a lot of numerical data. It is a spreadsheet software that allows you to store, organise, calculate and manipulate the data. It organises data in the form of rows and columns. Excel has a number of tools and functions that help in calculations and in analyzing data using formulas and tables. The various features of Excel are as follows:

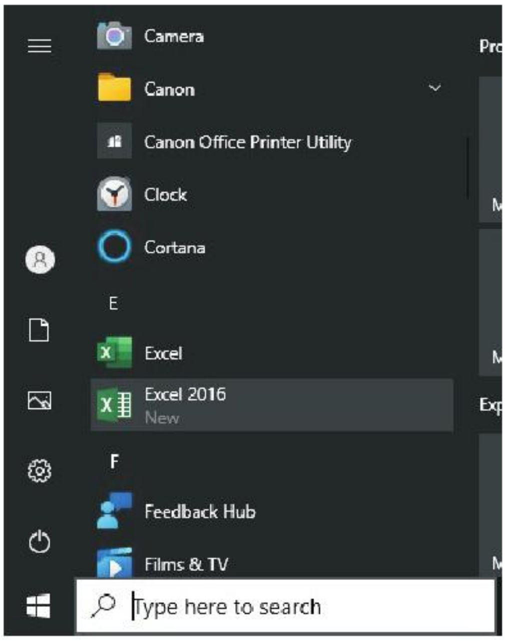
- It provides formulae for performing simple and complex calculations.
- It helps to search data quickly and replace it instantly.
- Data can also be viewed in graphical form like charts. It also helps us to compare data effectively.
- It also provides various formatting features to improve the appearance of data.

Starting Microsoft Excel

To start MS Excel 2016, follow these steps:

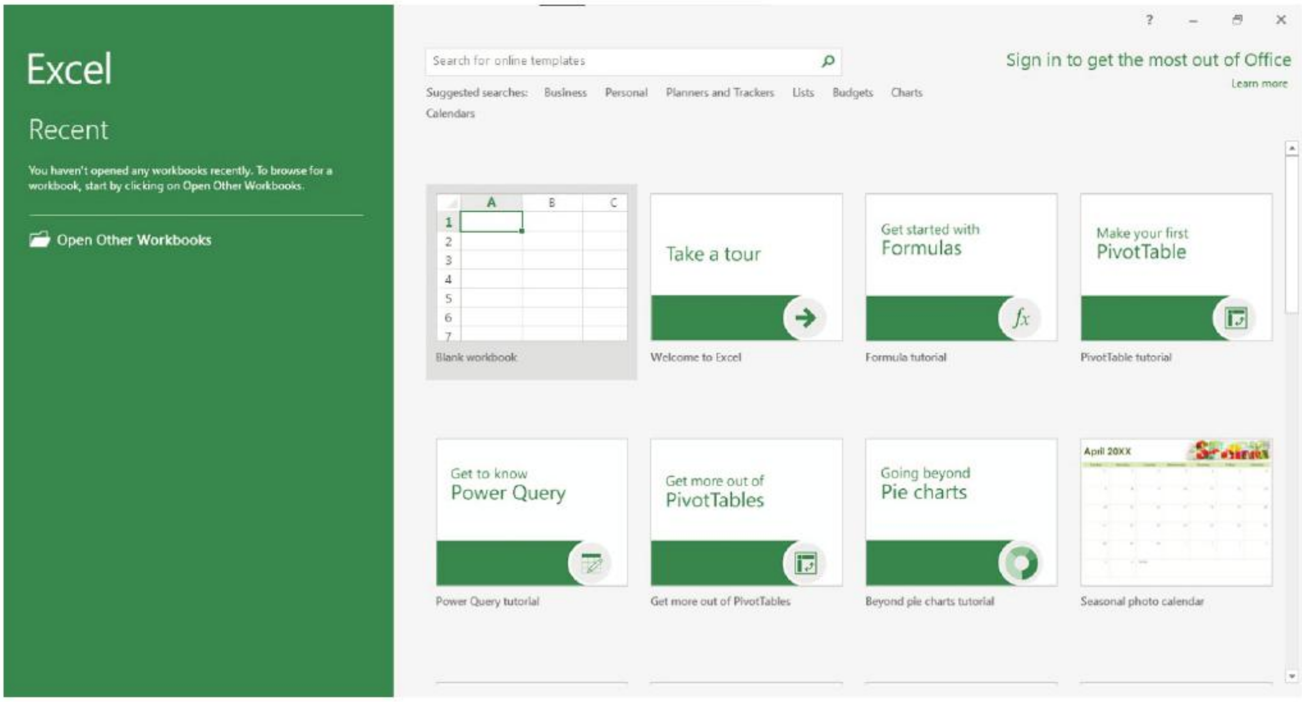
1. Click on the Start button.
2. Click on the Excel 2016.





When you open Excel 2016, the Excel Start Screen will appear. From here, you'll be able to create a new workbook.

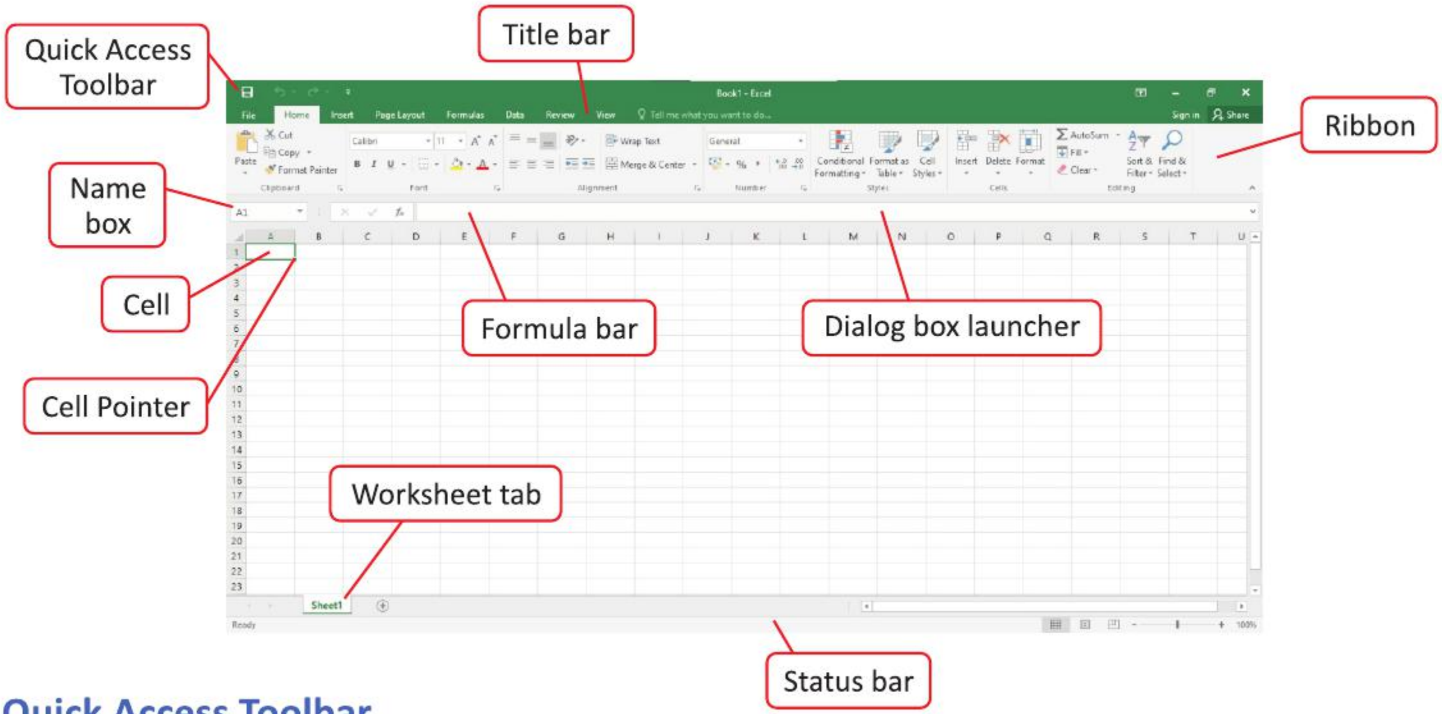
3. Click on the Blank workbook to create a new workbook.



Info Zone

You can also open Excel by typing Excel in the Search Box next to the Start button. Then, click on the Excel 2016 option.

Microsoft Excel window will appear named Book 1. The various parts of Excel 2016 window are as follows:



Quick Access Toolbar

It gives you quick access to commands you often use. By default, it includes Save, Undo and Redo commands. You can add more commands of your choice.

Title Bar

It is located at the top of the window. It displays the name of the workbook in which you are currently working. The right side of the title bar contains Minimize, Restore/Maximize and Close button.

Ribbon

It contains commands required to perform various tasks like formatting, sorting, etc. It has a number of tabs— File, Home, Insert, Page Layout, Formulas, Data, Review and View, each divided into groups. Each group consists of many commands.

Dialog Box Launcher

Some groups have a small arrow at the bottom right corner. This is the dialog box launcher. Clicking on this arrows, will open a dialog box in which you would see some more commands available in that group.

Workbook

A workbook is a collection of multiple worksheets stored under a single file name.

Worksheet

A worksheet is a work area made up of rows and columns where you enter and work with data. When you save an Excel worksheet, you are actually saving a workbook.

Row

It is an arrangement of cells in the horizontal direction. Each row on a worksheet is identified using a number. The rows are numbered from top to bottom in ascending order starting from 1, 2, 3 till 1,048,576. There are 1,048,576 rows in Excel 2016.

Column

It is an arrangement of cells in the vertical direction. Each column on a worksheet is identified using a unique label. The columns are labelled as A, B, C, ..., Y, Z followed by AA, AB, till XFD. There are 16,384 columns in Excel 2016.

Cell

A cell is the basic unit of a worksheet. It is formed by the intersection of a row and a column.

Cell Address

Every cell in a worksheet is identified using an address that corresponds to the column letter and the row number. Cell address is also known as cell reference.

Name Box

It displays the cell address or name given to a cell.

Formula Bar

It displays the contents of the current cell. It also displays the formula used in the cell.

It contains two buttons — Enter button and Cancel button.

- The Enter button works like the Enter key.
- The Cancel button can be used to cancel the data entry.

Cell Pointer

A cell pointer is a highlighted cell boundary  that indicates which cell is active at the moment.

Range of Cells

It is a group of cells adjacent to each other forming a rectangular shape.

Worksheet Tab

A workbook in Excel 2016 by default shows one worksheet, and its tab is by default named Sheet 1.

Scroll Bar

The Scroll bar appears only if you have data more than the screen can display at once. There are two scroll bars— the vertical bar on the right and the horizontal bar at the bottom.

Status Bar

It is present at the bottom of the window which displays the following:

a. Worksheet view options

- **Normal View :** By default, a worksheet opens in a normal view. This view shows you all the rows and columns in a worksheet.
- **Page Layout View :** It divides a worksheet into pages.
- **Page Break View :** It gives you an overview of a worksheet with page breaks, and tells you how many printed pages would be required to hold all the data in the worksheet. You can adjust the page breaks to increase or decrease the number of pages that will hold your data.

b. Zoom control

You can drag the slider left or right to decrease or increase the zoom. The number to the right of the slider reflects the zoom percentage.

Data Types

You can enter three types of data in an Excel worksheet. They are — Numbers, Text and Formulas.

Numbers or Numeric Data

It consists of numbers from 0 to 9 and symbols like +, -, *, /, $, %, etc. These are numeric data and can be used for calculations. By default, numbers are right-aligned in the cell.

Text

It consists of alphabets, numbers, spaces and special characters which are not used in calculations. By default, text data is left-aligned in the cell.

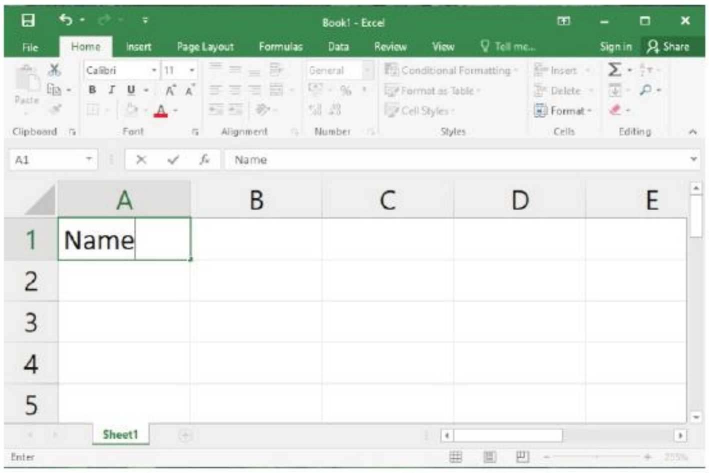
Formulas

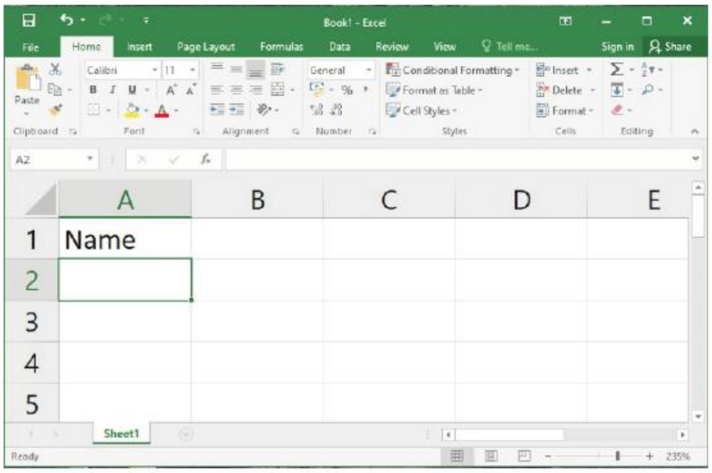
Formulas are the mathematical equations containing numbers, operators and cell address to perform calculations. Formulas always begin with equals to (=) sign.

Entering Data in a Worksheet

To enter data in a worksheet, follow these steps:

1. Click on the cell where you want to enter the data.
2. Type the data.
3. Click on the Enter button on the Formula bar or press the Enter key to accept the entry.





Moving Around in a Worksheet

Various methods of moving around in a worksheet are as follows:

Moves the cell pointer to	Keys
Up of the current cell	↑
Down of the current cell	↓
Left of the current cell	←
Right of the current cell	→
Next cell in a row	Tab
Previous cell in a row	Shift + Tab
One cell down in a row	Enter

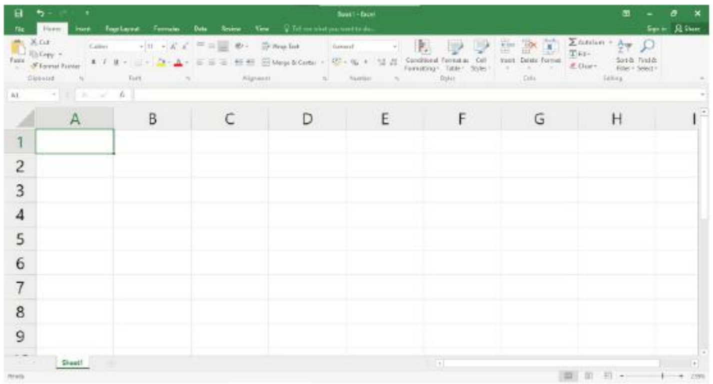
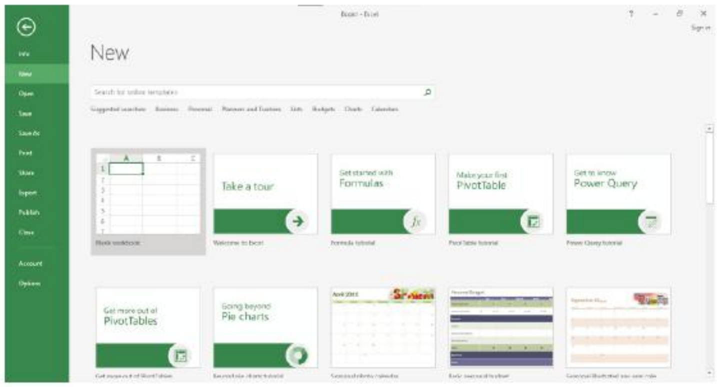
Moves the cell pointer to	Keys
First cell in the row	Howe
First cell in the worksheet	Ctrl + Home
Last used cell in a worksheet	Ctrl + End
One screen up	Page Up
One screen down	Page Down
One screen left	Alt + Page Down
One screen right	Alt + Page Up

Type the cell address in the Name box and press the Enter key, if you want to move the cell pointer to a particular cell in a worksheet quickly.

Creating a New Worksheet

To create a new workbook, follow these steps:

1. Click on the File tab. The Backstage view will appear.
2. Select New option.
3. Click on the Blank workbook.

A new blank workbook will appear on the screen.

Smart Tip

You can press the Ctrl+N keys to create a new workbook.

Auto Fill

Auto Fill is a feature in Excel that allows you to quickly create a series of numbers, dates or other items that follow a particular pattern.

You can generate a series by two ways:

1. Using Fill handle 2. Using Fill option

Using Fill Handle

Fill Handle is a small black square that appears when you place the mouse pointer at the lower-right corner of the selected cells . When you point to the Fill Handle, the cursor shape changes to a black plus sign .

To generate a series using Fill Handle, follow these steps:

1. Specify the initial value for the series in any of the cells.
2. Now drag the fill handle across the cells you want to fill.

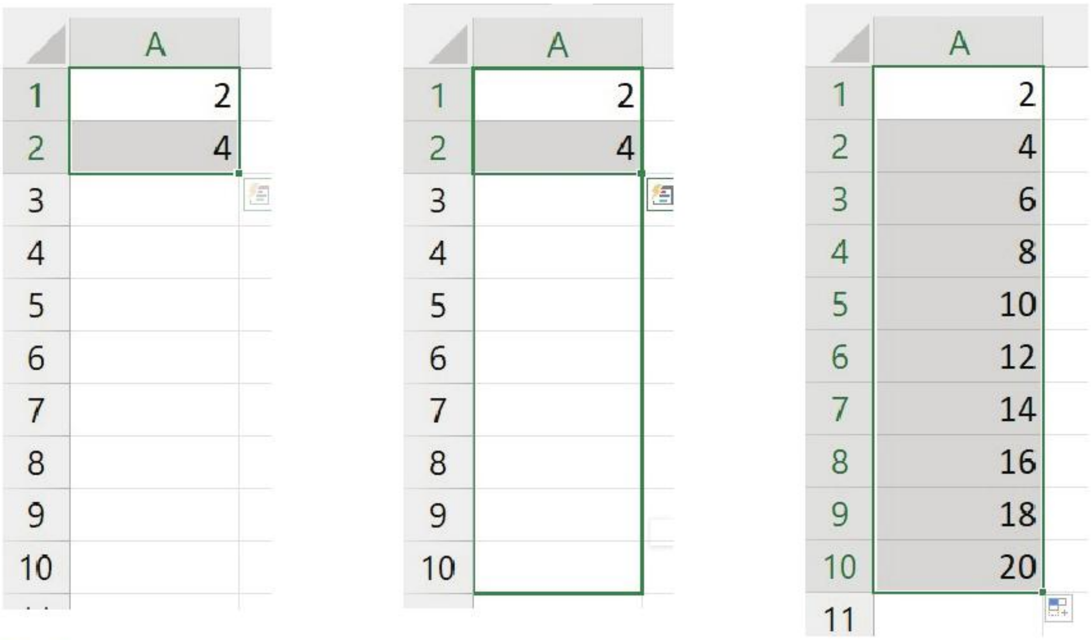
These types of series are predefined series. Predefined series include Sunday, Monday, ...; January, February, ...; and so on.

You can also fill the cells using the user-defined series. User-defined series include 10, 20, 30, ...; 7, 11, 15, ...; Jan, Apr, Jul, and so on.

To generate any user-defined series, follow these steps:

1. Specify the first two consecutive values of the series and select both the cells.
2. Drag the fill handle across the cells you want to fill.

Excel will pick up the pattern from the values entered and generate the rest of the series.



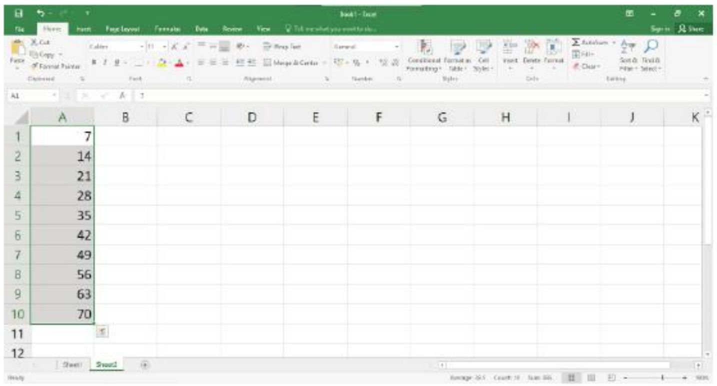
Using Fill Option

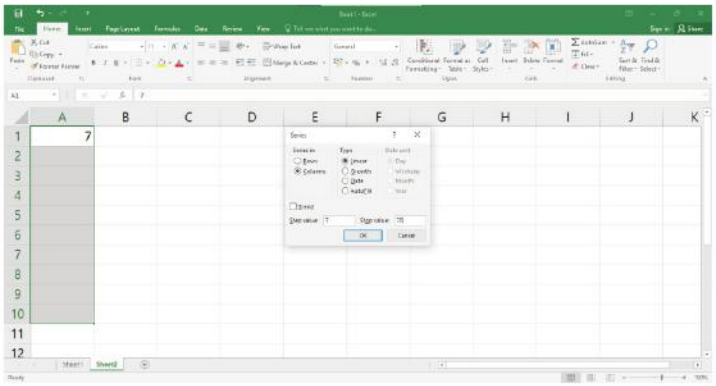
To generate a series using Fill option, follow these steps:

1. Specify the initial value for the series in any of the cell.
2. Select the range of cells to be filled in.
3. Click on the arrow next to the Fill option in the Editing group on the Home tab.
4. Select the Series option from the drop-down list. The Series dialog box appears.
5. Mention the Step value: and Stop value: as per your requirement in the dialog box.
6. Click on the OK button.

Info Zone

If you drag the fill handle in the downward or right direction, it fills the values in the increasing order, and if you drag the fill handle in the upward or left direction, it fills the values in the decreasing order.

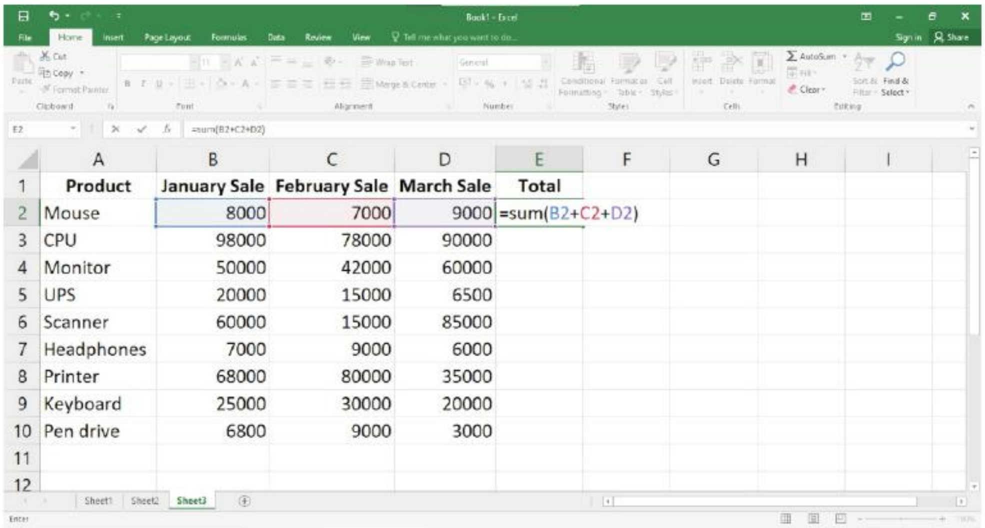




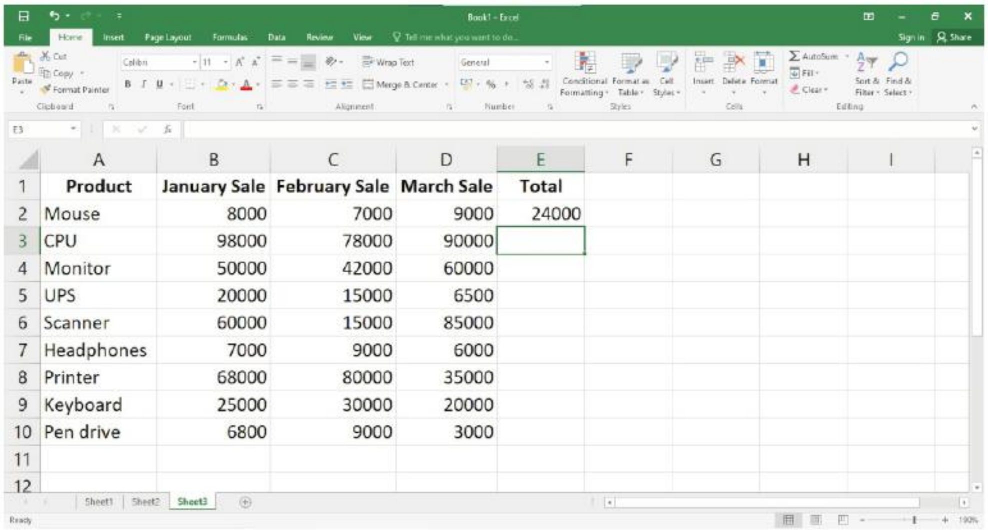
Performing Calculations

You can perform addition, subtraction, multiplication, and division of numbers using formulas or functions. The result of the calculation is displayed in the cell where the formula is entered. To perform calculation, follow these steps:

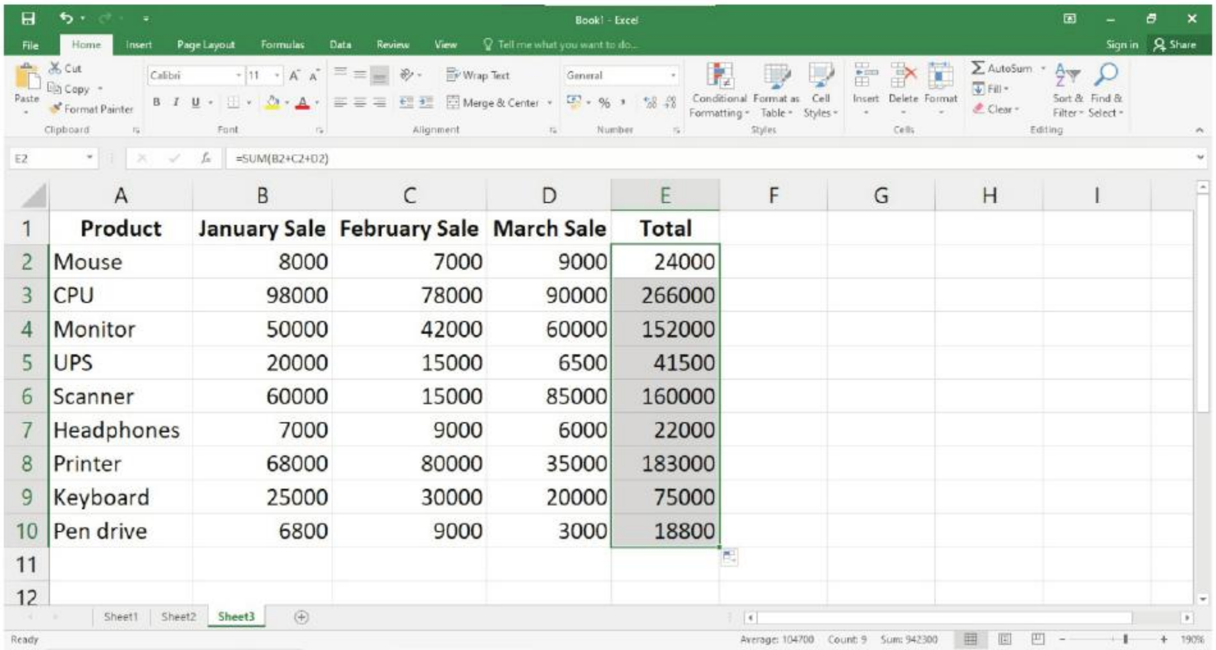
1. Type the data as shown and type the formula.



2. Press the Enter key.



3. Drag the fill handle across the cells you want to fill.



Working with Worksheets

When you open a new Excel workbook, three worksheets are added by default. You can add more worksheets to the workbook or delete worksheets that are no longer required from the workbook.

Renaming a Worksheet

You can change the name of a worksheet. To change the name of a worksheet, follow these steps:

1. Double-click on the name of the worksheet on the worksheet tab.
2. Type a new name and press the Enter key.

Adding a New Worksheet

You can add a new worksheet. To add a new worksheet at the end of the existing worksheet, follow these steps:

1. Click on the New sheet button present at the end on the worksheet tab.

   The new sheet will be inserted at the end of the existing worksheet.

To add a new worksheet between the existing sheets, follow these steps:

1. Select the worksheet before which you want to insert a new worksheet.
2. Click on the arrow next to the Insert option in the Cells group on the Home tab. A drop-down list is displayed.
3. Choose the Insert Sheet option.

A new worksheet is inserted before the selected worksheet.

Smart Tip

Press the Shift+F11 keys together to insert a new worksheet before the selected worksheet.

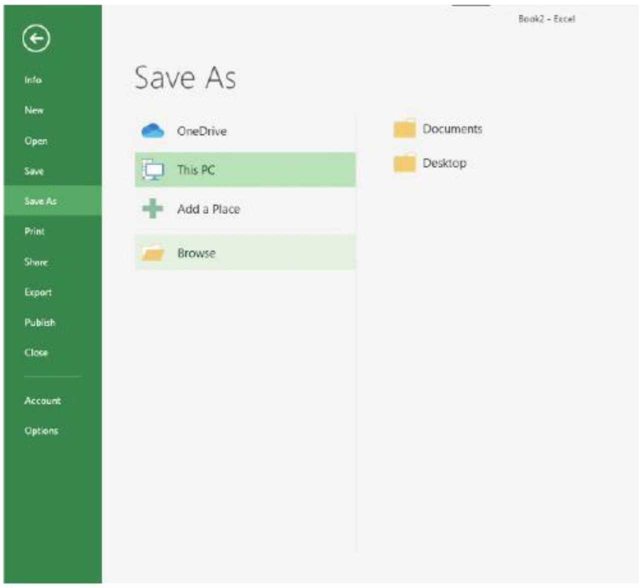
Info Zone

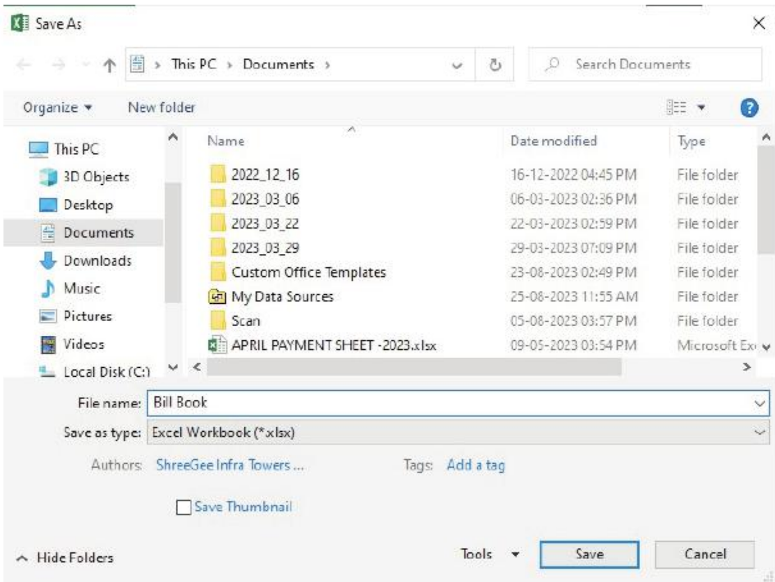
- Right-click on the worksheet tab and choose the appropriate option from the shortcut menu to rename or insert worksheets.
- When we add a number of worksheets, only a few number of worksheets can be seen on the screen. To switch between worksheets, you can use the Ctrl+PageUp and Ctrl+PageDown key combination.

Saving a Workbook

To save a workbook, follow these steps:

1. Click on the File tab.
2. Click on the Save option.
3. Click on the Browse button.
4. The Save As dialog box appears. Select the location where you want to save your workbook.
5. Type the file name.
6. Click on the Save button.





Smart Tip

- You can press the Ctrl+S keys to save the workbook in Excel 2016.
- You can also click on the Save button on the Quick Access Toolbar.
- The Save As command is used to create the copy of a workbook while keeping the original. When you use Save As, you will need to choose a different name and/or location for the copied version.

Let's Implement

A Tick (✓) the correct answer.

1. A ______________ is formed by the intersection of a row and a column.
   a. cell ◯ b. worksheet ◯ c. workbook ◯ d. cell address ◯
2. A ____________ is a collection of multiple worksheets stored under a single file name.
   a. cell pointer ◯ b. worksheet ◯ c. workbook ◯ d. cell ◯
3. There are ______________ rows in Excel 2016.
   a. 1,048,576 ◯ b. 1,049,576 ◯ c. 1,049,577 ◯ d. None of these ◯
4. By default, numbers are ______________ in the cell.
   a. left-aligned ◯ b. right-aligned ◯ c. top-aligned ◯ d. None of these ◯
5. ______________ displays the contents of the current cell.
   a. Formula bar ◯ b. Name box ◯ c. Worksheet ◯ d. Workbook ◯

B Fill in the blanks.

1. A ____________ is the basic unit of a worksheet.

2. There are _____________ columns in Excel 2016.
3. By default, text data is _____________-aligned in the cell.
4. The _____________ view divides a worksheet into pages.
5. A _____________ is a highlighted cell boundary that indicates which cell is active at the moment.

C Write True or False.

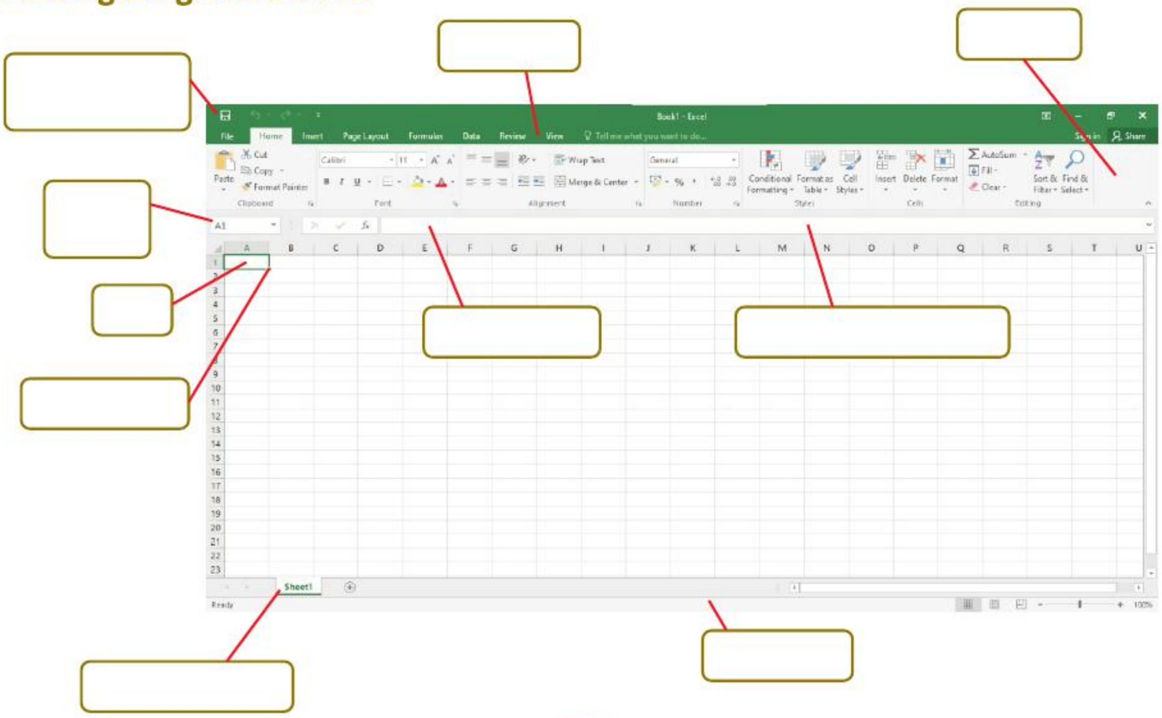
1. A row is an arrangement of cells in the vertical direction.
2. A workbook in Excel 2016 by default shows one worksheet.
3. All formulas always begin with plus (+) sign.
4. By default, a worksheet opens in a normal view.
5. Range of cells is a group of cells adjacent to each other forming a rectangular shape.

D Answer the following questions.

1. What is the difference between a worksheet and a workbook?
2. Define formula bar.
3. Name the different types of data entered in Excel.
4. Write the steps to rename a worksheet.
5. What is the use of Auto Fill?

Refreshment Corner

Label the figure given below.



- **Open Excel 2016 and create the following worksheet.**

W17

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	PRODUCT	JANUARY	FEBRUARY	MARCH	APRIL	MAY	JUNE	JULY	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER	DECEMBER	TOTAL
2	APPLES	2000	4000	4600	4500	1000	4500	4500	5443	4500	4500	6487	7654	53684
3	MANGOES	2000	5000	5600	3500	1200	4560	3456	5464	5600	4500	6453	6767	54100
4	PEARS	3000	3000	3400	4600	3400	5670	2345	5654	6700	3600	9820	7220	58409
5	PINEAPPLES	4000	2500	4500	5600	4500	6554	1234	2474	7600	4500	5360	1212	50034
6	APRICOTS	2000	4500	5600	6700	4600	2399	2332	8530	2300	2345	2323	2323	45952
7	BANANAS	2400	5700	3400	2300	4500	2345	3434	4500	4500	6540	5467	4545	49631
8														

To find the TOTAL use =B2+C2+D2+E2+F2+G2+H2+I2+J2+K2+L2+M2 formula in cell N2. Then, drag the fill handle across the cells you want to fill. Finally, save it by the name 'Sales Report'.

RACKING YOUR BRAIN

Life Skills and Values

Your friend is trying to perform calculation in a table created by him/her in MS Excel. But, he is unable to complete the task. Then, what will you do?

Just Have a Discussion

Communication

Discuss the various parts of Excel 2016 window in the class in detail.

Teacher's Note

Students should be given enough practice time to work on Excel 2016, so that they familiarize themselves with tabs and buttons of Excel 2016.

7

INTERNET BASICS

Outlines

- Internet Related Terms
- Recent Development
- Introduction to Robotics

Internet is a network of networks in which users can do anything on any one computer and if something is missing and they have the permission, they can get the information from any other computer. The U.S. Department of Defence laid the foundation of the Internet roughly 30 years ago with a network called ARPANET. But the general public didn't use the Internet much until after the development of the World Wide Web in the early 1990.

Info Zone

ARPANET served only computer professionals, engineers and scientists, who knew their way around its complex workings.

Internet Related Terms

- **Browser :** Software used to find, retrieve, view, and send information over the Internet.
- **Download :** To copy data from a remote computer to a local computer.
- **Upload :** To send data from a local computer to a remote computer.
- **E-mail (electronic mail):** It is used for exchange of messages by telecommunication. You can also send images and sound files, as attachments.
- **Filter :** Software that allows targeted sites to be blocked from view. Examples : X-Stop, AOL@School.

- **Home Page :** The beginning 'page' of any site.
- **HTML (HyperText Markup Language) :** The language used to create web pages for use on the World Wide Web.
- **HTTP (HyperText Transport Protocol) :** The set of rules for exchanging files (text, images, sound, video and other multimedia files) on the World Wide Web.
- **Hypertext :** Generally any text that contains 'links' to other text.
- **Search Engine :** It provides links to pages that contain the object of your search.
- **TCP/IP (Transmission Control Protocol/Internet Protocol):** It is the communication language or protocol of the Internet.
- **URL (Uniform Resource Locator) :** In the Internet address, the URL indicates which area of the Internet will be accessed.
- **WWW (World Wide Web) :** It includes all the resources and users on the Internet that are using the Hypertext Transfer Protocol (HTTP).

Introduction to Robotics

Robotics technologies deal with automated machines that can take the place of humans in dangerous environments or manufacturing processes or resemble humans in appearance, behaviour and/or cognition.

Info Zone

- 'Robotics' means the science or study of the technology associated with the design, fabrication, theory and application of robots.
- Robots are any machine that does work on its own or automatically.

Artificial Intelligence : It is the ability of a computer or other machine to perform those activities that are normally thought to require intelligence.

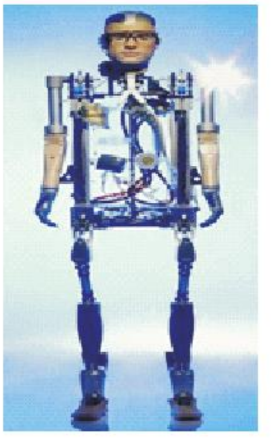
Robot Mechanics : Scientists are working hard on building a robot that can perform a number of complex movements as well as utilize a variety of sensors.

Robots in Industry : Robots are ideal for doing repetitive or dangerous tasks. Around 90% of robots are used in factories, with half of these being used in the automobile industry.

Robots and robotic arms are frequently used for:

- Car manufacturing
- Military — Bomb disposal, weapons, army surveillance
- Medical — Surgery, X-Rays, life support
- Space — Shuttles, International Space Station, Mars rovers

Bionic Man



The Bionic Man is the world's first robot human. He walks, talks and he has a heart beating but he's not human, he's the world's first fully bionic man.

The bionic man's hand is made by Touch Bionics with a wrist that can fully rotate and motors in each finger. The hand's grasping abilities are impressive, but still drops drinks sometimes.

FemiSapien

FemiSapien is a female humanoid robot. It can respond to sight, sound and touch and can be programmed with a sequence of movements. FemiSapien is the latest in the line of sophisticated, walking, talking, personality packed robots. Intelligent and interactive, FemiSapien speaks her own language called 'emotish' which consists of gentle sounds and gestures. There is no remote required. You can interact with her directly and she responds to your hand gestures, touch and sound.

Info Zone

- Roboticists Rich Walker and Matthew Godden of Shadow Robot Co. in England led the assembly of the bionic man.

Features: Three modes of interaction.

1. **Learning Mode :** can be programmed with a sequence of movements.
2. **Responsive Mode :** make her respond to walking commands and also you can interact with her as she reacts to your sounds or watch her perform comedy scenes with you or another robot.
3. **Attentive Mode :** you can activate actions such as autonomous wandering, blowing kisses, poses, holding hands, dancing, belching and she will even engage you in 'conversation'.

Recent Development

3G : 3G is shorthand for '3rd generation' and refers to a networking standard in cell phone technology that is capable of providing high-speed data service to mobile devices. As 3G wireless networks became more widespread, you could connect to and use the Internet at practical speeds, which surpassed those of the previous generation of mobile phone technology (called 1X).

4G : 4G stands for '4th generation' mobile data protocol. Using a 4G smartphone on Verizon's 4G LTE (Long Term Evolution) network means you can download files from the Internet up to 10 times faster than with 3G. With 4G LTE, using the web from your phone becomes as pleasurable as using it from your home computer.

Info Zone

- Verizon's 3G network paved the way for a world that's almost forgotten phones that were once used merely for voice calls and text messaging.

Kirobo : Kirobo is Japan's first robot astronaut, developed by the University of Tokyo and Tomotaka Takahashi. A twin to Kirobo, named Mirata was created with the same characteristics. Mirata will stay on Earth as a backup crew member. The word 'kirobo' itself is a portmanteau of 'kiboo' which means 'hope' in Japanese and the word 'robo' used as a generic short word for any robot.

Skype : Skype is for doing things together, whenever you're apart. Skype's text, voice and video make it simple to share experiences with the people that matter to you, wherever they are. You can use Skype on whatever works best for you—on your phone or computer or a TV with Skype on it. You can even try out group video, with the latest version of Skype.

Google Glass : Google Glass is a type of wearable technology with an optical head-mounted display (OHMD). It was developed by Google with the mission of producing a mass-market ubiquitous computer. Google Glass displays the information in a smartphone-like hands-free format.

Info Zone

Google Glass wearers can communicate with the Internet via natural language voice commands.

Android : One of the most widely used mobile OS these days is ANDROID. Android is a software bunch comprising not only operating system but also middleware and key applications. Android Inc was founded in Palo Alto of California, U.S. by Andy Rubin, Rich Miner, Nick Sears and Chris White in 2003. Later, Android Inc was acquired by Google in 2005. After the original release, there have been a number of updates in the original version of Android.

Let's Implement

A Tick (✓) the correct answer.

1. To send data from a local computer to a remote computer.
   a. Upload ◯ b. Download ◯
   c. Hyperlink ◯ d. Hypertext ◯
2. The set of rules for exchanging files on the World Wide Web.
   a. HTM ◯ b. HTTP ◯
   c. URL ◯ d. WWW ◯
3. In the Internet address, it indicates which area of the Internet will be accessed.
   a. HTTP ◯ b. HTML ◯
   c. WWW ◯ d. URL ◯
4. It is a type of wearable technology with an optical head-mounted display.
   a. Robotics ◯ b. Google Glass ◯
   c. Android ◯ d. Skype ◯
5. One of the most widely used mobile OS these days.
   a. Windows ◯ b. 3G ◯
   c. Android ◯ d. 4G ◯

B Fill in the blanks.

1. _____________ is a software used to find, retrieve, view and send information over the Internet.
2. ___________ is the language used to create web pages for use on the World Wide Web.
3. _____________ is the beginning 'page' of any site.
4. _____________ is communication language or protocol of the Internet.
5. _____________ technologies deal with automated machines that can take the place of humans.

C Write **True** or **False**.

1. Upload means to copy data from a remote computer to a local computer. ▭
2. Hyperlink is generally any text that contains 'links' to other text. ▭
3. Skype is for doing things together, whenever you're apart. ▭
4. Google chrome displays the information in a smartphone—like hands-free format. ▭
5. Robots are ideal for doing repetitive or dangerous tasks. ▭

D Match the following.

1. 

3.

5. 

(a) Bionic man

(b) FemiSapien

(c) Kirobo

(d) Skype

(e) Google Glass

(f) Android

2. 

4. 

6. 

E Answer the following questions.

1. Define search engine.
2. What is TCP/IP?
3. What is URL?
4. What is WWW?
5. What is Google Glass?

Refreshment Corner

- **Find these missing words in the given maze.**

Upload, Filter, Blocked, HTML, HTTP, Protocol, Internet, Google, Skype, URL, TCP

F	I	L	T	E	R	S	D	F
A	N	B	L	O	C	K	E	D
H	T	M	L	B	J	G	U	A
P	E	T	A	F	L	O	P	S
P	R	O	T	O	C	O	L	K
S	N	T	C	P	A	G	O	Y
D	E	D	F	U	R	L	A	P
H	T	T	P	A	S	E	D	E

Practical Session

Experiential Learning

- **Browse on the Internet, Search for the topic 'National Animal of India'. Go through various links and download it.**

  Steps to be followed:

  - Browse the Internet.
  - For this, open the Google website.

- **Now type 'National Animal of India' on Google and search. Visit various links shown on the search pages.**

- **Download the matter that is suitable for you.**

RACKING YOUR BRAIN

Life Skills and Values

Your friend is not able to browse the internet then what will you do? Justify your answer.

Just Have a Discussion

Communication

Discuss the different uses of Internet in the class in detail.

Teacher's Note

Explain the various terms of Internet to the students in detail.

Scan Me

8

AI IN DAILY LIFE

Outlines

- Google Lens
- Google Search
- Navigation
- Gmail
- Recommendations in Apps

AI is a trending technology. Its popularity is increasing day-by-day, like other technologies. There are numerous AI-enabled applications that you use on a daily basis. Let us discuss them in detail.

Google Lens

Google Lens is an image recognition tool, developed by Google. It lets you search for what you see. It uses visual analysis powered by the neural network to bring up information related to an object. A neural network is a computer architecture that has multiple layers. It is used to understand and solve the problems using artificial intelligence, just like the neurons in our brain.

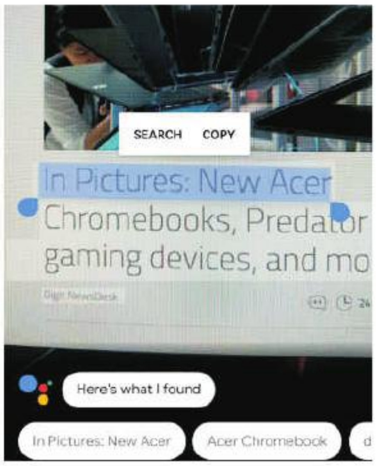
Google Lens

Applications of Google Lens

The following are various applications of Google Lens:

Text Translation : By using Google Lens, you can translate the text to the language of your choice in real-time.

The Translation feature of Google Lens is helpful when you find difficulty in reading the text, like signboards on the road, which are written in a language that you cannot speak or understand. This is the type of situation you mostly face when you go abroad. Here, Google Lens comes as a handy tool. You can place your phone on the text and translate the selected words into any language within a matter of few seconds. Google Lens can make your language and communication easier.



Smart Text Selection : You can point your smartphone's camera at a particular text, highlight it within the Google Lens, and copy it to your phone or search for the same.

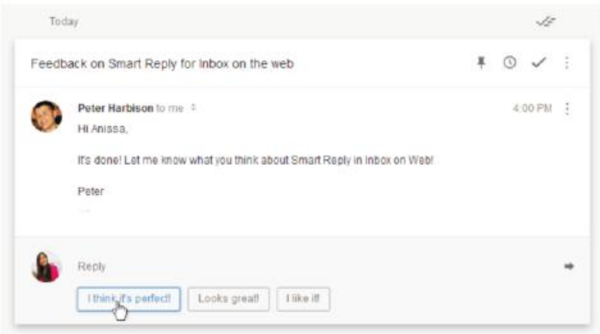
Search Your Surroundings : If you point your camera around you, Google Lens can identify your surroundings and provide the information related to it.

Know the Details of an Image : You can capture a picture and use Google Lens to search for the information about the captured picture. For example, you can capture a picture of the monument that you want to visit and search for all the information related to it using the Google Lens. In the same way, you can take a picture of the dress and search for that article while shopping. It is a helpful tool to get quick information about plants, flowers, or other images.

Assist in Understanding Concepts : If you find difficulty in understanding any concept, just scan the text using Google Lens and it will provide you with all the information about the concept. For example, you can get help with mathematical and chemical equations, etc.

Gmail

Gmail uses AI to filter your emails into different categories, like Personal, Primary, Social, Promotion, Spam, etc. Working in the background, AI scans the emails received in your account and transfers them into the correct folder category.

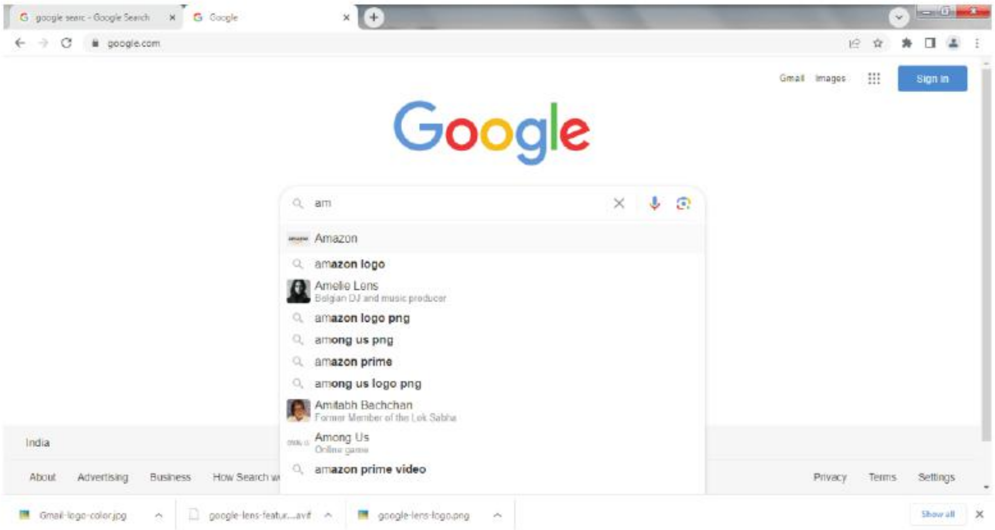


Smart Reply in Gmail

Smart Reply is a useful feature of Gmail. It uses AI to scan received emails and suggest quick responses according to the test of those emails. To write an instant reply, the user simply has to click on the test suggestions displayed at the bottom of the email.

Google Search

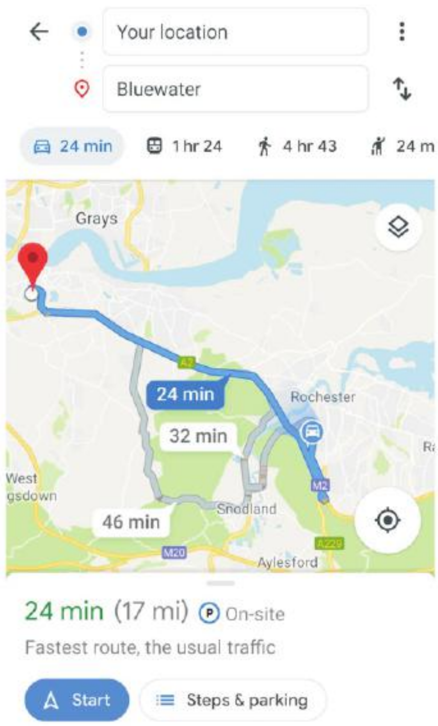
Google is widely used to search for information on the internet. Every time you search for something on Google, artificial intelligence is working behind the scenes to generate more relevant responses to your query. AI helps the Google Search engine understand exactly what a user is searching for and then deliver personalized results based on what it knows about him or her.



Recommendations in Apps

Most of the streaming apps like Netflix, Disney Hotstar, YouTube, etc. use AI to recommend shows and movies based on the viewer's choice. Various music apps like Spotify, play music based on the user's listening habits. E-commerce apps like Amazon, Myntra and Flipkart, also use AI for product recommendations.

Navigation



Navigation apps like Google Maps and Apple Maps, use AI to update a commuter about traffic congestion in a particular area and about the estimated time to reach a particular destination. While you are travelling in your car, AI calculates the average speed of your car, and the distance between your current location and destination. To update you about traffic congestion, it takes note of the number of active mobile connections in an area where the traffic is expected and relays the information on your mobile app.

COMPUTER ETIQUETTES

1. Keep your face directly forward and close your eyes.
2. Slowly move the head forward. Try to touch your chin to the chest.
3. Move your head as far back as is comfortable. Do not strain your neck.
4. This is one round. Practice 10 rounds.

Let's Implement

A **Tick (✓) the correct answer.**

1. _______________ is a navigation app.

   a. Google Maps ◯ b. Netflix ◯ c. Gmail ◯

2. _______________ apps provides movie recommendations as per your choice.

   a. Google engine ◯ b. Myntra ◯ c. Netflix ◯

3. Google Lens _______________ translate text.

   a. Can ◯ b. Cannot ◯ c. Both a & b ◯

4. AI helps the _______________ engine understand exactly what a user is searching for and then deliver personalised results.

   a. Google Search ◯ b. Gmail ◯ c. Netflix ◯

5. _______________ is a music app.

   a. AI ◯ b. Spotify ◯ c. Flipkart ◯

B **Fill in the blanks.**

1. _____________ is an image recognition tool.
2. Using AI, the _____________ feature of Gmail scans the received email and suggests a quick reply.
3. _____________ is a streaming app.
4. _____________ apps use AI for product recommendations.
5. Google Lens helps to get quick information about plants, ___________ or other images.

C **Write True or False.**

1. Google Lens cannot help you identify the images. ☐
2. A neural network is a computer architecture that has multiple layers. ☐
3. Google Lens can translate the text to the language of your choice. ☐
4. Gmail uses AI to filter your emails into different categories. ☐
5. We cannot translate the text using Google Lens. ☐

D Answer the following questions.

1. What is Google Lens?
2. Write any two uses of the Google Lens?
3. Describe the role of AI in Gmail.
4. Explain the role of AI in Google Search.
5. How does AI help when you are travelling in your car?

Refreshment Corner

- **Draw and colour the logos of Google Lens and Google Maps in the space given below.**

Art Integration

RACKING YOUR BRAIN

Life Skills and Values

Your friend is trying to use the Google Lens but unable to use it. What will you do?

Just Have a Discussion

Communication

Discuss the different applications of Google Lens in the class in detail.

Teacher's Note

Demonstrate the use of AI-enabled applications that we use to the students in detail.

9

Scan Me

MORE ON SCRATCH

Outlines

- Using Sensing Blocks
- Comparing Values
- Bouncing the Ball Up and Down
- Making a Variable
- Using Mathematical Operators
- Generating Random Numbers
- Counting Length of a Word
- Applying Condition

Using Sensing Blocks

The Sensing blocks palette has light blue colour-coded blocks, which are used to sense the keyboard input while executing the script. The block prompts the user to type the input using the keyboard, and the block stores the information or input entered by the user using the keyboard. The question appears in a speech bubble on the screen. The program waits until the user types his response in the Input box and presses the Enter key or clicks on check mark. Let us create small script by using the sensing blocks and execute it:

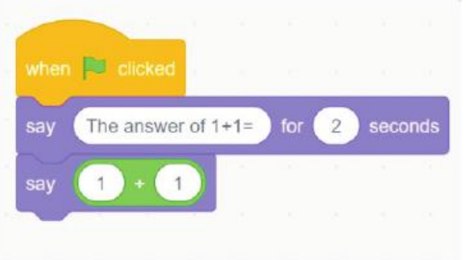
- Delete the Cat sprite from the stage.
- Click on the Choose a Sprite icon and add the Visual Programming Language. Choose the appropriate costume, Kai-a, for the sprite from the Costumes palette.
- Select the Stage thumbnail on the left and click on the Choose a Backdrop icon, import the School backdrop on the stage.
- Create the script by dragging and placing the three blocks as shown in the figure 8.1..
- The block enables the user to type the input with the help of a keyboard.
- Drag and drop the block on the text field of the block. Click on the Green Flag (Go) button.
- The message 'Where should I go?' gets displayed on the stage along with a text box.

- Type the answer 'To my friend's place' in the text box. Press the Enter key or click on the check mark present on the answer text box.
- The answer will be shown in the speech bubble on the stage for a specified number of seconds.

Using Mathematical Operators

All the blocks in the Operators blocks category are round in shape. They are light green colour-coded blocks. They are used to solve mathematical equations. These blocks can be easily placed in the value box of any other block. Let us perform calculations in a Scratch project using these blocks:

- Open a new file in the Scratch window. Delete the default Cat sprite. Click on Choose a Sprite button. Select the Animals category from the Choose a Sprite dialog box.
- Select the Duck sprite from the options.
- Drag the block from the Events block category and drop it in the Scripts area. Drag the blocks from the Looks blocks category and snap them together underneath the block.



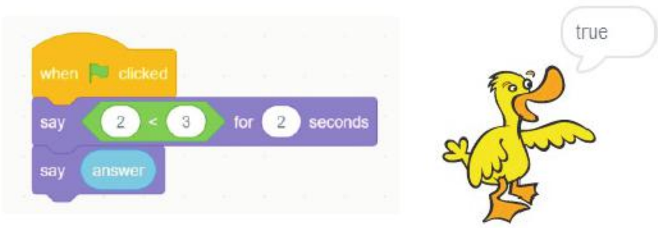
- Type the message 'The answer of 1+1= in place of 'Hello!' in the block.
- Click and drag the block from the Operators blocks category and drop it inside the block.
- Type the numbers in the block as shown in the figure.
- Click on the Green Flag (Go) button to run the script and observe the sprite calculating the given numbers.

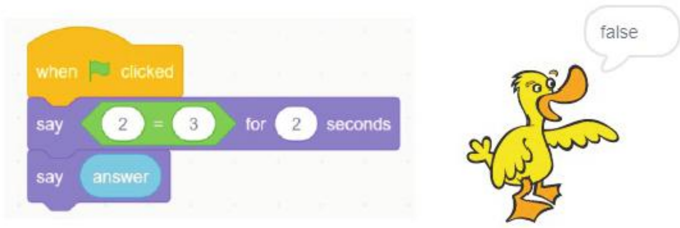
- Change the numbers and observe the sprite displaying a change in the result.
- Similarly, you can use the other blocks to perform subtraction, multiplication, and division from the Operators block category.

Comparing Values

The Operators block palette also includes blocks to compare the numbers. In Mathematics, you use less than (<), greater than (>), or equal to (=) sign to compare two numbers. Similarly, in Scratch, the (less than) block reports true if the first value is less than the second value, the (equal to) block reports true if the two numbers are equal, and (greater than) block reports true if the first value is greater than the second value. These blocks are known as comparison blocks. Let us make use of the comparison blocks:

- Create a script by dragging the blocks as shown in the figure 8.4 (a) and type the numbers 2 and 3 in the (less than) block to check whether 2<3. Click on the Green Flag (Go) button to run the script. You will get the answer as true.





- Likewise, create another script using the (equal to) block to check whether 2 is equal to 3 as shown in the figure 8.4 (b) and execute it. You will get the answer as false.

Generating Random Numbers

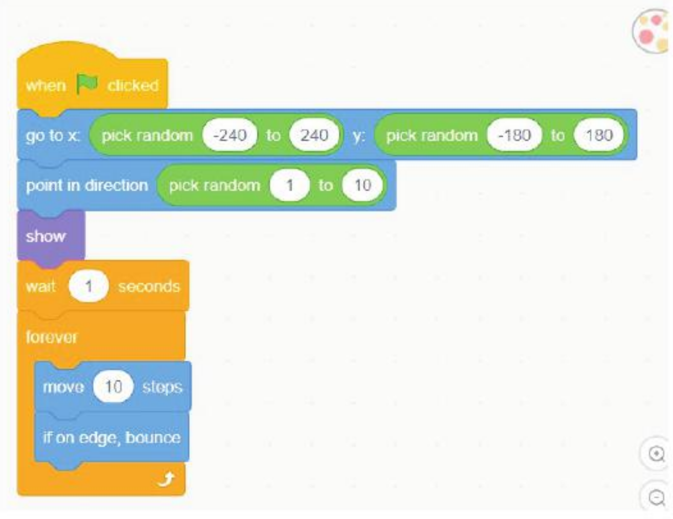
You can also generate a random number from a specified range by using the block from the Operators blocks category. Let us see how the sprite itself choose the random numbers with the help of this block.

- Place the blocks in the Scripts area and change the values as shown in the figure.

- Click on any block to execute it. Notice that the sprite will show the random numbers between 1 and 50.

Bouncing the Ball Up and Down

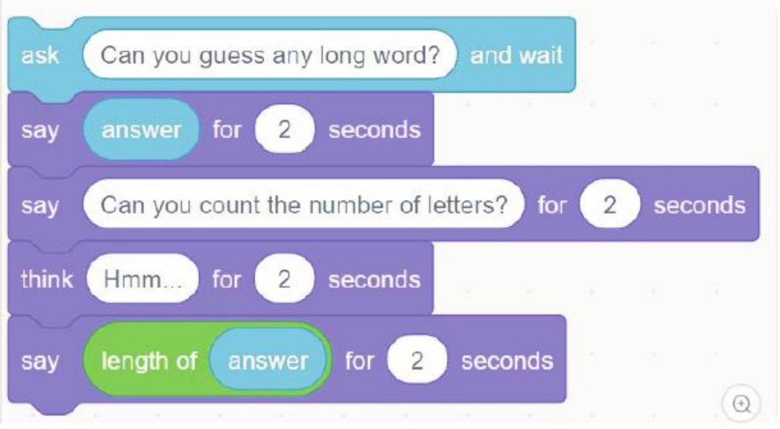
To bounce a ball all around the stage, you can use the block along with the other blocks. Create a new Scratch project using the Beachball sprite and the blocks as shown in the figure.



Counting Length of a Word

Scratch blocks provide the facility to count the letters that are used to form a word. For example, the word Encyclopedia has 12 letters. Let us create a script to display the total number of letters:

- Open a new file in Scratch by clicking on the File > New.
- Select the Stage thumbnail and click on the Choose a Backdrop icon and add Bedroom3 from the Indoors category. Click on OK.
- Select the Code tab and drag the block from the Sensing blocks palette and drop it on the Scripts area.
- Type the text 'Can you guess any long word?' in the text field of the block.
- Drag the block from the Looks blocks category in the Scripts area. Pick the block from the Sensing blocks category and place it where the 'Hello!' text is written.

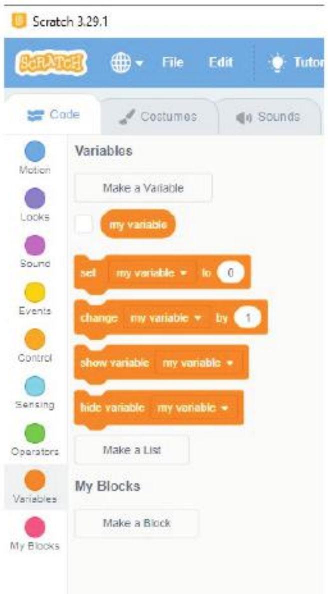


- Again, place the block underneath the script. Change the text 'Hello!' to 'Can you count the number of letters?'
- Add the block.
- Now, again place the block in the Scripts area and add the block from the Operators blocks category to the text box of block as shown in the figure.
- Now, drag the block from the Sensing blocks category and place it on top of the text box in the block.
- Click on any block to run the script and observe the sprite displaying the length of the word.

Making a Variable

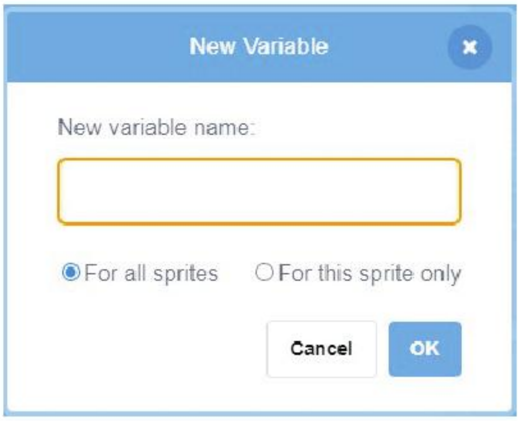
You might have observed that while crossing the different levels of a computer game, the score of the player either increases or decreases. It is displayed on the screen at the end of the game. To store these score values, variables are used.

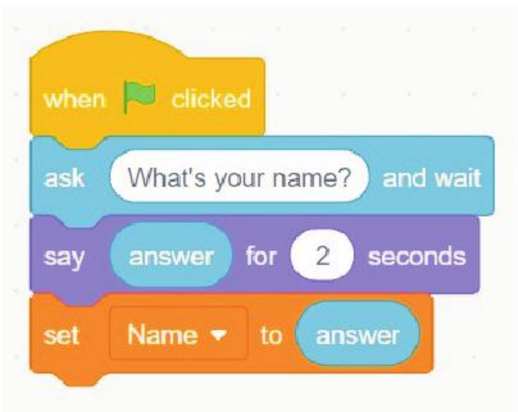
A variable is a placeholder. It is used to store a value. It can hold one value at a time. Values can be either number of Strings (text).



In Scratch, the Variables blocks category is used to create variables in a project. Variables are represented by blocks, shaped like elongated circles uniquely labelled by the user. Let us learn how to create a variable:

- In the Variables blocks palette, click on the Make a Variable block. The New Variable box appears.
- Type any suitable name like 'Name' for the variable in the Variable name text box. Select a button specifying whether the variable is For all sprites or For this sprite only. Click on OK.
- The 'Name' block will be added under the Variables blocks menu.





- Now, drag the block as shown in figure to create a script, using the different block palettes.
- Click on the Green Flag (Go) button to run the script.
- Type your name with the help of the keyboard in the answer box displayed on the stage, and press the Enter key.

- The sprite will speak the answer in a speech bubble and it will be stored under the variable name on the top-left corner of the stage.

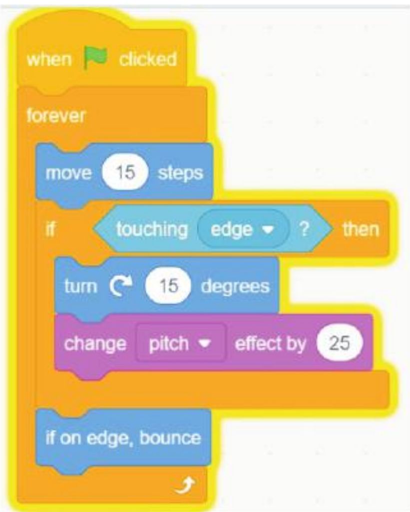
Applying Condition

To put a check on any problem, a condition is to be applied. It allows the program to select an action, based upon the user's input. Scratch provides various blocks enabled with conditional programming concepts, like 'if-then', 'forever-if' and 'if-else'. These are the conditional blocks present in the Control block palette.

In the block, the script will execute only if the condition evaluates to be true. All the diamond-shaped blocks of the Sensing blocks palette can be placed as a condition inside the Control blocks.

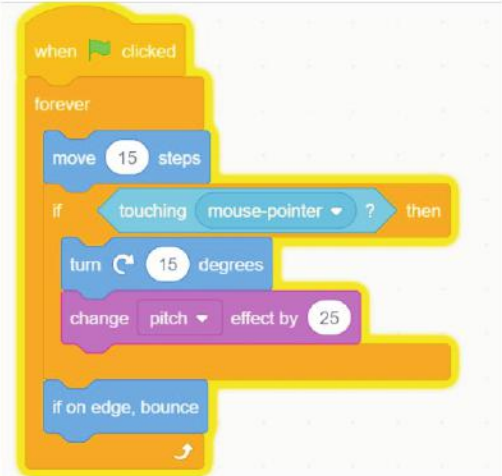
Let us observe them in a project:

- Open a new file in Scratch. Import the backdrop Soccer from the Sports category as shown in the figure.



- Delete the Cat sprite and add Soccer Ball sprite from the Sports category on the stage. Reduce its size by putting value in the Size box.
- Drag the block to the Scripts area. Place the block beneath it.
- Add the block from the Motion category. Change the value to 15.
- Place the conditional block beneath the block.
- Drag the block from the Sensing blocks palette, and place it over the value box in the block to check a condition and take the decision.
- Select the edge option from the drop-down menu of the block lets you check whether the sprite is touching the edge of the stage or mouse pointer.

- Drag the block from the Motion category and place it beneath the block. Change the value from 15 to 5.
- Drag the block from the Sound category and place it under the block.
- Drag the block from the Motion category and keep it outside the block.
- Click on the Green Flag (Go) button and observe the ball moving all around the stage.
- Now, select the mouse-pointer option from the drop-down list of the block as shown in the figure.



- Again click on the Green Flag (Go) button and run the script to observe the change.
- Observe that the ball is moving and changing its colour only if you place the mouse pointer on the ball. Click on the Stop button to halt the execution.

COMPUTER ETIQUETTES

1. Close your eyes tightly for 3-5 seconds.
2. Now open them for 3-5 seconds.
3. Repeat this 7 or 8 times.

Let's Implement

A **Tick (✓) the correct answer.**

1. The block to add two numbers are located under the ____________ block menu.

   a. Operators ◯ b. Sensing ◯

   c. Both a & b ◯ d. None of these ◯

2. A ____________ is a placeholder that stores a value.

a. variable ◯ b. Operator ◯

c. block ◯ d. None of these ◯

3. The ____________ blocks palette is used to create variables in the scratch project.

a. Sensing ◯ b. Operators ◯ c. Variables ◯ d. Blocks ◯

4. The ____________ blocks category holds the blocks that are used to apply conditions, like if-else.

a. Control ◯ b. Motion ◯ c. Sensing ◯ d. None of these ◯

B Fill in the blanks.

1. The blocks in the ____________ blocks category are round in shape.
2. The sensing blocks palette has ____________ colour-coded blocks.
3. Values can be either of number ____________.
4. Variables can hold ____________ value at a time.

C Write **True** or **False**.

1. You cannot perform mathematical operations in Scratch. ☐
2. All the blocks in the Operators blocks category are light blue in colour. ☐
3. Scratch blocks provide the facility to count the letters that are used to form a word. ☐
4. You can create variables in a Scratch project. ☐

D Answer the following questions.

1. What is the use of variables?
2. What are comparison blocks?
3. What are conditional blocks?
4. Can we count the letters with Scratch blocks?

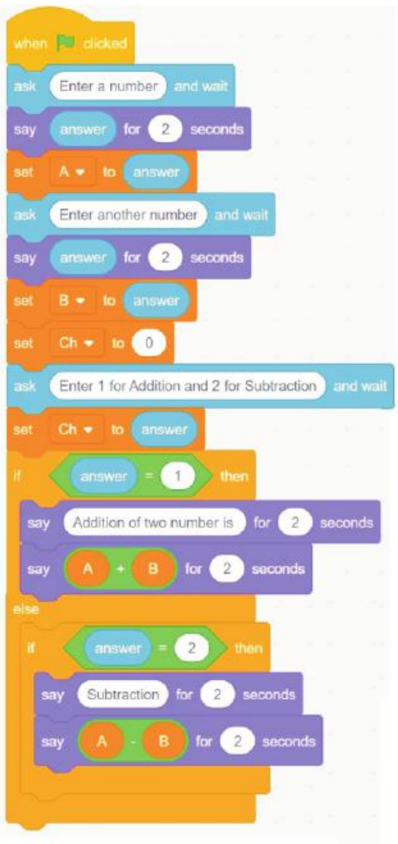
Refreshment Corner

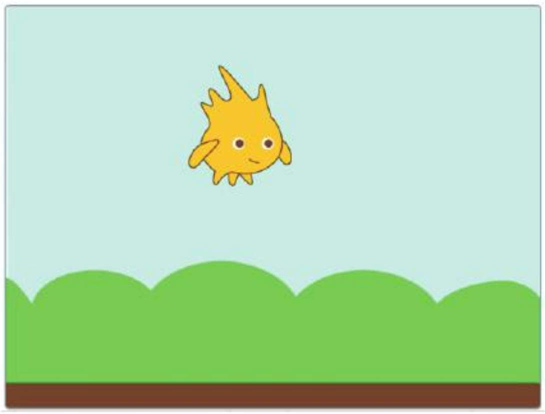
- **Draw and colour the picture of any sprite of Scratch3 in the space given below.** Art Integration

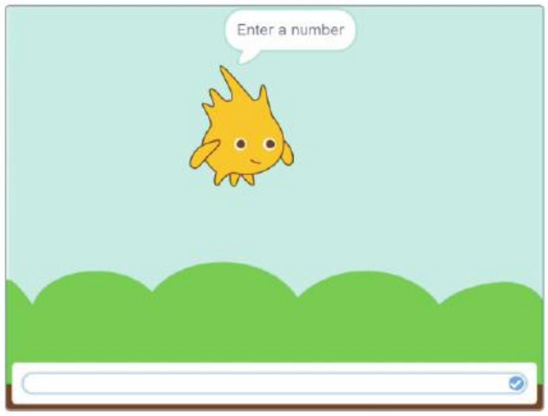
Practical Session

Create a scene given below and script in Scratch by following these steps.

- Click on the Choose a Backdrop button and select the Blue Sky backdrop.
- Delete the existing sprite from the stage. For this, click on the Cross (✗) sign on the sprite in the Sprite List box.
- Click on the Choose a Sprite button and select the Gobo sprite.
- Create the given script and execute it.







RACKING YOUR BRAIN

Life Skills and Values

If your teacher tells you to improve your project, then what will you do— ignore what he/she says, copy from your friend and show to your teacher or listen to your teacher and improve your project?

Just Have a Discussion

Communication

Discuss the different Operators blocks in the class in detail.

Teacher's Note

Demonstrate the use of various Sensing blocks to the students in detail.

QUICK REVIEW 2

Tick (✓) the correct option.

1. ________ is not a category under the Animation effects.
   - a. Entrance effects
   - b. Enlivening effects
   - c. Both a and b
   - d. None of these
2. A ________ is a collection of multiple worksheets stored under a single file name.
   - a. cell pointer
   - b. worksheet
   - c. workbook
   - d. cell
3. In the Internet address, it indicates which area of the Internet will be accessed.
   - a. HTTP
   - b. HTML
   - c. WWW
   - d. URL
4. AI helps the ________ engine understand exactly what a user is searching for and then deliver personalised results.
   - a. Google Search
   - b. Gmail
   - c. Netflix
   - d. None of these
5. The Audio option is present in the ________ group on the Insert tab.
   - a. Media
   - b. Create Graphic
   - c. Both a and b
   - d. None of these
6. There are ________ rows in Excel 2016.
   - a. 1,048,576
   - b. 1,049,576
   - c. 1,049,577
   - d. None of these
7. The set of rules for exchanging files on the World Wide Web.
   - a. HTM
   - b. HTTP
   - c. URL
   - d. WWW
8. ________ is a navigation app.
   - a. Google Maps
   - b. Netflix
   - c. Gmail
   - d. Myntra
9. ________ effects control the manner in which the object exits during the slide show.
   - a. Entrance
   - b. Exit
   - c. Both a and b
   - d. None of these
10. By default, numbers are ________ in the cell.
    - a. left-aligned
    - b. right-aligned
    - c. top-aligned
    - d. None of these

TEST PAPER 1

A **Fill in the blanks.**

1. Integrated circuits were used in _______________ generation of computers.
2. The _______________ of the Taskbar shows the programs that are open.
3. The _______________ tool is used to include effects.
4. Shapes option is in the _______________ group.

B **Write True or False.**

1. Blaise Pascal invented the method of logarithm. [ ]
2. Small pictures on the desktop are called graphics. [ ]
3. You cannot change unit from percentage to pixel. [ ]
4. When you click on the inserted WordArt, the Format tab appears. [ ]

C **Answer the following questions.**

1. What do you mean by digital computers?
2. What is the use of This PC?
3. What is Effects tool?
4. What is WordArt?

D **Name the following.**

1. Process of loading of operating system in memory _______________
2. GUI operating system _______________
3. Any two icons _______________
4. Two Windows 10 apps _______________

E **Identify the pictures and name the generations of these computers.**

_______________ _______________ _______________

TEST PAPER 2

A Fill in the blanks.

1. _____________ effects control the manner in which the object is introduced into the slide during the slide show.
2. There are _____________ columns in Excel 2016.
3. Values can be either number of _____________ .
4. _____________ is the communication language or protocol of the Internet.

B Write True or False.

1. Upload means to copy data from a remote computer to a local computer. ☐
2. A neural network is a computer architecture that has multiple layers. ☐
3. Scratch blocks provide the facility to count the letters that are used to form a word. ☐
4. Wheel is an Emphasis effect. ☐

C Answer the following questions.

1. What do you mean by SmartArt Graphics?
2. What is the use of Auto Fill?
3. Write any two uses of the Google Lens?
4. Can we count the letters with Scratch blocks?

D Match the following.

1. 

2. 

3. skype

4. 

5. 

6. 

(a) Bionic man
(b) FemiSapien
(c) Kirobo
(d) Skype
(e) Google Glass
(f) Android